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Agentic AI Auditability & Assurance White Paper 2026

A Lifecycle Evidence Guide for Audit, Assurance, and Enterprise AI Governance

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Relationship to the Agentic Lifecycle Governance Industry Series

The Agentic Lifecycle Governance Industry Series is fixed as a 3+3 structure: three core industry white papers and three practitioner guides. The Global AI Compliance White Paper 2026 defines the compliance object layer for agentic lifecycle governance. This paper defines the audit evidence object layer. Guide 1 will later translate this paper into audit-ready technical implementation planning. Guide 2 will later translate the Global AI Compliance White Paper 2026 and this paper into compliance operating-model planning. The later insurability white paper will address insurability. This R4 package prepares candidate source for review; it does not publish or replace those later assets.

Executive Thesis

AI agent auditability cannot be built on logs alone. An agentic system becomes auditable only when its lifecycle work can be reconstructed through responsibility-linked evidence chains.

Agentic AI Auditability is the ability to reconstruct, test, and evidence agentic lifecycle work across authority, responsibility, tools, outcomes, exceptions, and remediation. This definition is author synthesis in this paper grounded in audit evidence language, AI governance guidance, provenance/logging concepts, privacy/evidence-retention guidance, and GAIC's MRO object layer.

Source Note

Big Four sources are used as market context only. Professional audit and assurance sources support terminology, evidence concepts, internal-control language, and assurance boundaries. AI governance/public guidance sources support governance context.

Evidence/provenance/observability/logging sources support the logs-versus-evidence distinction. Privacy sources support evidence minimization, selective disclosure, and retention tension. GAIC source truth supports MROs, Validation Lab boundaries, RCCS-M/ALCS context, and companion-paper boundaries. Inline source IDs in the candidate source map to the package source register and package citation map.

Boundary Note

This candidate source is not an audit standard, certification, legal compliance proof, assurance opinion, regulator-approved method, procurement recommendation, vendor ranking, Big Four endorsement, audit body endorsement, MPLP requirement, or MPLP industry-standard claim. AARM is a proposed readiness model, not a score, benchmark, certification, assurance result, or vendor assessment.

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0. Executive Summary

Candidate source status: R4 internal candidate source

Chapter purpose: State the auditability problem, the object shift, the evidence-chain answer, the AARM bridge, and the non-standard boundary.

Reader question: Why does agentic AI need a new auditability frame?

Key Claims

- AI agent auditability cannot be built on logs alone.
- Agentic systems become auditable only when lifecycle work can be reconstructed through responsibility-linked evidence chains.
- Auditability shifts the review object from model output alone to lifecycle work across authority, responsibility, tools, outcomes, exceptions, privacy treatment, and remediation.
- GAIC MROs can be translated into audit evidence objects and evidence requests.
- AARM is a proposed readiness model for describing auditability readiness; it does not issue assurance, certification, or legal compliance proof.

Source Grounding Note

Market context is grounded in BF-01, BF-02, and BF-05. Audit evidence language is grounded in AUD-01 and AUD-02. AI governance context is grounded in AI-01 and AI-08. Observability context is grounded in EVID-02. MROs, Validation Lab boundaries, and companion-paper boundaries are GAIC-derived source truth.

Author Synthesis Note

Agentic AI Auditability, Agentic Audit Object, Audit Evidence Chain, lifecycle-responsibility-linked agent work, and AARM are author synthesis in this paper. They are not presented as externally issued audit standards or certification schemes.

Candidate Prose

AI agent auditability cannot be built on logs alone. Logs matter. Traces matter. Workflow histories, tool-call records, telemetry, and incident records all help reconstruct technical activity. But agentic auditability asks a different question: not merely what happened inside a system, but whether the work can be reviewed as lifecycle responsibility.

The shift is subtle and material. Traditional AI governance often begins with the model, the dataset, the output, the control environment, and the policy framework. Those layers remain necessary.

An agentic system, however, does not only produce outputs. It delegates work, calls tools, transfers tasks across agents, accepts or disputes outcomes, escalates exceptions, stores evidence, crosses vendor or processor boundaries, and may require remediation. A log can record that a tool call occurred. By itself, it usually does not establish who authorized the action, which human role owned the work, whether the agent acted inside a delegated authority boundary, whether the outcome was accepted, what exception occurred, or whether remediation closed.

This paper defines **Agentic AI Auditability** as the ability to reconstruct, test, and evidence agentic lifecycle work across authority, responsibility, tools, outcomes, exceptions, and remediation. This is an authored definition. It is grounded in professional audit evidence language, AI governance guidance, provenance concepts, observability/logging documentation, privacy guidance, and GAIC's existing lifecycle responsibility object layer.

It is not a legal definition, audit standard, assurance engagement requirement, certification

criterion, or regulator-approved compliance test.

The central contribution of this paper is the Audit Evidence Object layer. The Global AI Compliance White Paper 2026 defined Missing Regulatory Objects for agentic lifecycle governance. This paper translates those objects into auditability terms: evidence requests, walkthrough structures, responsibility maps, evidence partitioning, privacy-aware disclosure, third-party review boundaries, and readiness levels.

The Agentic Audit Object is the unit of review. The Audit Evidence Chain is the responsibility-linked structure that connects lifecycle work to authority, role, tool action, evidence pointer, accepted outcome, exception state, and closure state.

This matters because audit and assurance readers increasingly face AI systems whose behavior cannot be understood through model cards, output samples, policy attestations, or raw telemetry alone. Big Four and professional sources already discuss AI assurance, trusted AI, responsible AI, audit transformation, governance, risk, controls, and agentic AI adoption. Those discussions create the market context for this paper.

This paper does not claim that any firm endorses this framework. Instead, it offers an object-model layer adjacent to those discussions: a disciplined way to ask what must exist before agentic work can be reconstructed.

The paper's recurring distinction is simple:

- Logs record activity.
- Observability explains system behavior.
- Audit evidence supports responsibility review.
- Responsibility-linked evidence chains make agentic lifecycle work reconstructable.

The distinction does not make logs unimportant. Logs and traces are often evidence ingredients. They may identify time, sequence, tool, service, user, or execution path. But audit evidence requires more than event capture. It requires relevance, reliability, sufficiency in context, relationship to review objective, and linkage to responsibility. For agentic systems, that means linking activity to lifecycle work units, human roles, agent roles, delegated authority, accepted outcomes, exception handling, privacy treatment, and remediation closure.

This paper also introduces the **Agentic Auditability Readiness Model (AARM)**. AARM describes whether an agentic system is unobservable, log-visible, trace-linked, evidence-

structured, auditability-ready, or assurance-ready for planning purposes. AARM is deliberately bounded. It is not a score, benchmark, certification, audit opinion, assurance conclusion, legal compliance proof, regulator approval, procurement recommendation, or vendor ranking. Its purpose is to help organizations and reviewers discuss readiness before they overstate assurance.

For enterprises, this paper prepares the path to two practitioner guides. Guide 1 will translate auditability into technical architecture for CIO, CTO, platform, security, and engineering teams. Guide 2 will translate auditability into compliance operating-model design for CCO, governance, legal, risk, internal audit, and policy-to-evidence teams. This paper defines what audit-ready means; the later guides will address how to build and govern toward it.

For audit and assurance firms, this framework can be used as a discussion and readiness structure. It can support evidence request framing, walkthrough scoping, object-model review, and internal method conversations. It does not replace professional methodology, professional judgment, engagement acceptance, independence rules, attestation criteria, assurance standards, or audit procedures. The paper is useful only if its boundary is kept visible.

Executive Orientation Table

Paper element	What it means	Source / synthesis status	Boundary
Agentic AI Auditability	Ability to reconstruct, test, and evidence agentic lifecycle work across authority, responsibility, tools, outcomes, exceptions, and remediation	author synthesis in this paper grounded by AUD-01, AUD-02, AI-01, AI-08, EVID-01, EVID-02, and GAIC-SOURCE	Not audit standard, legal proof, or assurance opinion
Agentic Audit Object	Reviewable lifecycle work object for agentic systems	author synthesis in this paper derived from GAIC MROs and provenance/audit evidence concepts	Not mandatory schema
Audit Evidence Chain	Responsibility-linked chain connecting work unit, authority,	author synthesis in this paper	Not a certification

	role, tool, evidence, outcome, exception, privacy treatment, and closure		criterion
MRO-to-evidence mapping	Translation of GAIC MROs into audit evidence objects and evidence requests	GAIC-derived plus audit/control language	MROs are not legal mandates
AARM	Proposed readiness model for auditability	author synthesis in this paper grounded by R4B/R1 and audit/governance sources result	Not score, certification, or assurance

Cross-Links

- See Appendix A for evidence request categories.
- See Appendix C for the full MRO-to-audit-evidence mapping.
- See Appendix D for the AARM readiness matrix.
- See Appendix F for boundary language.

Boundary Note

This chapter does not claim this paper is published, final, sealed, regulator-approved, audit-body-endorsed, Big-Four-endorsed, legally sufficient, certifying, or assurance-producing.

1. Scope, Audience, and Non-Audit-Standard Boundary

Candidate source status: R4 internal candidate source

Chapter purpose: Define who this paper is for, what it does, what it does not do, how sources are used, and how GAIC/MPLP/Validation Lab relate to the draft.

Reader question: Who should use this paper, and what authority must it not claim?

Key Claims

- This paper is a lifecycle evidence guide and auditability readiness framework.
- This paper is not an audit standard, assurance opinion, certification, legal compliance proof, regulator approval, procurement recommendation, vendor ranking, or Big

Four/audit body endorsement.

- Big Four sources are market context only.
- GAIC is the source for MROs, AARM planning, Validation Lab boundary, and companion-paper scope.
- MPLP may be discussed as one optional lifecycle protocol path, never as required or industry-standard.

Source Grounding Note

Assurance and attestation boundaries use AUD-03 and BOUND-03. Certification and conformity-assessment boundaries use BOUND-01 and BOUND-02. Governance role context uses AUD-07. Big Four AI assurance context uses BF-02 and BF-03 only as market context.

Author Synthesis Note

This paper's scope, Audit Evidence Object layer, Agentic Audit Object, Audit Evidence Chain, and AARM are author synthesis. They are draft framework constructs, not externally adopted professional requirements.

Candidate Prose

This paper is written for readers who need to understand whether agentic AI work can be reconstructed. Its primary audience includes audit and assurance professionals, internal audit teams, technology risk teams, AI governance committees, enterprise control owners, CIOs, CTOs, CCOs, CROs, legal/privacy stakeholders, and platform leaders responsible for agentic systems. Its secondary audience includes advisory teams, AI assurance practitioners, risk engineering teams, agent runtime builders, procurement and vendor-risk teams, and standards or policy readers exploring agentic AI governance.

The paper's scope is deliberately narrow. It does not try to restate all model governance, responsible AI, cybersecurity, privacy, or enterprise risk management. Those fields remain necessary. This paper focuses on the lifecycle evidence layer that becomes necessary when AI systems act through agents, tools, memory, delegation, multi-agent handoffs, and remediation workflows. The question is not simply whether a model behaved acceptably. The question is whether the lifecycle work can be reconstructed across authority, responsibility, tools, outcomes, exceptions, and closure.

This paper is best understood as a lifecycle evidence guide. It defines the proposed object layer needed for audit-ready agentic work, maps GAIC MROs into audit evidence objects, proposes

evidence request categories, provides walkthrough and partitioning patterns, introduces AARM readiness levels, and preserves boundaries around privacy, third-party validation, and assurance. It is a draft architecture for auditability, not a professional standard.

The source hierarchy matters. Professional audit and assurance sources provide language for evidence, controls, assurance boundaries, internal audit, attestation, and professional limits. AI governance sources provide context for risk management, accountability, monitoring, documentation, and human oversight. Evidence, provenance, observability, and log-management sources help distinguish raw event data from responsibility-linked evidence. Privacy sources ground the tension between reviewability, minimization, selective disclosure, and retention. Big Four sources provide market context: they show that AI assurance, trusted AI, audit transformation, and agentic AI are active enterprise topics. They do not endorse this paper.

GAIC source truth plays a different role. The Global AI Compliance White Paper 2026 created the MRO object layer and the broader Agentic Lifecycle Governance framing. This paper uses those GAIC-derived objects as its internal foundation. When this paper discusses MRO-01 through MRO-16, RCCS-M/ALCS context, the Evidence-Based Validation Pattern, Validation Lab boundary language, or companion-paper sequencing, it is relying on GAIC source truth, not external audit standards.

MPLP should be handled with care. MPLP may be described as one protocol path whose lifecycle responsibility semantics can help express evidence objects. This paper does not require MPLP, does not rank MPLP above other approaches, does not claim MPLP is an industry standard, and does not claim MPLP proves compliance, auditability, assurance readiness, or enterprise readiness. Guide 1 may later discuss implementation paths, but this paper must remain implementation-neutral.

Validation Lab is also boundary-sensitive. GAIC describes Validation Lab as a non-certifying evidence adjudication example. This paper may use that boundary as an example of third-party evidence review without certification. It must not describe Validation Lab as a certification body, conformity assessment body, regulator, audit firm, assurance provider, legal authority, or proof of compliance.

The safest reading of this paper is this: it provides a structured way to ask what evidence must exist before agentic AI work can be meaningfully reviewed. It does not determine whether that evidence is sufficient for a particular audit, assurance engagement, legal analysis, regulatory

process, insurance underwriting decision, procurement decision, or board conclusion. Those determinations belong to qualified professionals operating under their own methods, scopes, independence requirements, criteria, and legal obligations.

Scope Table

Area	This paper does	This paper does not do	Source / boundary grounding
Auditability	Defines proposed lifecycle evidence objects and readiness questions	Issue audit opinions or audit procedures	AUD-01, AUD-02, AUD-03
Assurance	Frames readiness for possible assurance planning	Provide assurance conclusion or engagement standard	AUD-03, BOUND-03
Certification	Distinguishes validation from certification	Certify systems or define certification criteria	BOUND-01, BOUND-02
Legal/privacy	Frames evidence/privacy tension	Provide legal advice or legal compliance proof	PRIV-01 to PRIV-05, AI-09
Big Four context	Uses public materials as market context	Claim endorsement, adoption, or need	BF-01 to BF-05
GAIC/MRO	Uses MROs as GAIC-derived object layer	Claim MROs are law or external standards	GAIC-SOURCE
MPLP	Treats MPLP as optional protocol path	Require MPLP or claim industry-standard status	GAIC-SOURCE
Validation Lab	Gives non-certifying evidence adjudication example	Claim certification, assurance opinion, or regulator approval	GAIC-SOURCE, BOUND-01, BOUND-02

Cross-Links

- Appendix F contains reusable boundary language.

- Chapter 12 expands the validation/certification boundary.
- Chapter 15 addresses professional use by audit and assurance firms.

Boundary Note

This chapter intentionally repeats non-claim language because this paper's usefulness depends on not overstating its authority.

2. Why AI Agent Auditability Is Different from Model Governance

Candidate source status: R4 internal candidate source

Chapter purpose: Explain why model governance is necessary but insufficient for agentic auditability.

Reader question: What changes when AI systems act through agents, tools, memory, delegation, and multi-agent workflows?

Key Claims

- Model governance remains necessary for agentic systems, but it does not fully reconstruct lifecycle work.
- Agentic AI introduces delegated authority, tool action, role separation, handoffs, accepted outcomes, exception states, and remediation closure.
- Auditability must therefore move beyond model/output review into lifecycle responsibility review.

Source Grounding Note

Big Four market context appears in BF-01, BF-04, and BF-05. AI governance context appears in AI-01, AI-02, AI-04, AI-05, and AI-08. Internal audit context appears in AUD-06. GAIC source truth provides MRO and RCCS-M/ALCS lifecycle-object context.

Author Synthesis Note

The claim that "model governance is necessary but insufficient" for agentic auditability is author synthesis in this paper grounded in the shift from model/output governance to lifecycle responsibility objects.

Candidate Prose

Model governance is not obsolete. It remains part of responsible AI practice. Organizations still need to understand model purpose, data, evaluation, limitations, monitoring, robustness, bias, security, human oversight, and change management. Public AI governance frameworks and professional guidance continue to provide important language for risk management, accountability, documentation, monitoring, controls, and internal audit review.

The limitation is not that model governance is wrong. The limitation is that agentic systems create work that is not reducible to model behavior. An agent may plan a sequence, invoke a tool, pass work to another agent, store state, retrieve memory, request human confirmation, act under delegated authority, trigger an external workflow, or remediate an exception. A model output may be only one event in a broader lifecycle.

For auditability, that broader lifecycle matters. If an agent drafts a message, the model output may be reviewed. If an agent sends the message through a CRM tool, updates a customer record, triggers a refund, routes a contract, or changes a production configuration, the relevant review object is no longer only the output. It is the work unit: who initiated it, what authority permitted it, what agent and tool executed it, what evidence was retained, who accepted the result, what exception occurred, and how any remediation closed.

This creates an object gap. Traditional governance may document the model, evaluate outputs, approve use cases, and monitor performance. Agentic auditability must also document authority boundaries, role mappings, tool-action evidence, responsibility transfer, evidence partitioning, privacy treatment, and closure. These are lifecycle objects, not merely model attributes.

The distinction becomes clearer when human oversight is considered. A human-in-the-loop control may be meaningful only if the loop is tied to an identifiable responsibility state. Who was the human? What role did they occupy? What were they asked to confirm? Was the confirmation advisory, blocking, or final? Was it tied to the delegated authority boundary? Did it create an accepted outcome state? Did it leave evidence that another reviewer can reconstruct? Without these objects, "human oversight" may be real operationally but weak evidentially.

Agentic systems also complicate responsibility because agent roles are not human roles. Human-like names, personas, and task labels may help interfaces, but they can blur governance. An "analyst agent" is not the same thing as an analyst. It is a bounded execution role with tool permissions, instructions, constraints, evidence obligations, and escalation paths. A human or organizational role still owns intent, authority, acceptance, and remediation.

The point of this paper is to place model governance beside lifecycle evidence architecture. Model governance asks whether the model and its use are governed. Agentic auditability asks whether the agentic work can be reconstructed. Both are needed. The second cannot be inferred from the first.

Table 1: Traditional AI Audit vs Agentic AI Auditability

Traditional AI audit focus	Agentic auditability focus	Missing lifecycle object	Evidence needed	Source / synthesis note	Boundary
Model documentation	Lifecycle work reconstruction	Agentic Audit Object	Work unit ID, scope, evidence chain, accepted outcome	Mixed: AI-01, AI-08, GAIC-SOURCE, author synthesis in this paper	Not a new audit standard
Output review	Accepted outcome review	Accepted outcome state	Reviewer role, acceptance/rejection/dispute state, evidence pointer	GAIC-derived + AUD-01/AUD-06	Acceptance is not legal proof
Monitoring	Authority and exception monitoring	Authority drift record	Baseline authority, observed action, escalation, closure evidence	AI-02, EVID-04, GAIC-SOURCE	Drift is not automatically a legal breach
Access control	Delegated authority review	Delegated authority boundary	Scope, condition, expiry, tool	AI-08, AUD-06, GAIC-SOURCE	IAM permission is not full business

			action,		authority
			escalation		
			Human role,		
	Lifecycle	Responsibility	agent role,	AUD-04,	Walkthrough is
Control testing	responsibility	map and	tool record,	AUD-05,	not formal audit
	walkthrough	evidence chain	partitioned	EVID-01	procedure
			evidence		

Table note: Mixed source-grounded, GAIC-derived, and author-synthesis table. It is not a certification table, audit procedure, procurement tool, or vendor comparison.

Cross-Links

- Chapter 3 defines the audit object shift.
- Chapter 5 defines the Agentic Audit Object Model.
- Appendix C maps MROs to audit evidence objects.

Boundary Note

This chapter does not attack model governance or claim model governance is useless. It argues that agentic auditability requires an additional lifecycle responsibility layer.

3. The Audit Object Shift: From Model Output to Lifecycle Work

Candidate source status: R4 internal candidate source

Chapter purpose: Define the proposed review object for agentic auditability.

Reader question: What exactly should audit stakeholders reconstruct?

Key Claims

- Agentic auditability requires a shift from model/output review to lifecycle work review.
- The Agentic Audit Object is a proposed review object for lifecycle-responsibility-linked agent work.
- Lifecycle-responsibility-linked agent work connects intent, authority, role, agent action, tool action, evidence, outcome, exception, privacy treatment, and closure.

Source Grounding Note

Audit evidence language is grounded in AUD-01 and AUD-02. Governance context is grounded in AI-01 and AI-08. Provenance concepts are grounded in EVID-01. MRO source truth is GAIC-derived.

Author Synthesis Note

Agentic Audit Object and lifecycle-responsibility-linked agent work are author-synthesis constructs. They are not claimed as existing professional standards, legal categories, or regulator-defined objects.

Candidate Prose

An audit object is the thing being reviewed. In many AI governance conversations, the object is a model, dataset, output, use case, control, system, or policy. Those objects remain important. But agentic systems create another review object: lifecycle work.

Lifecycle work is not a single output. It is the chain of actions and decisions by which an agentic system receives intent, acts under authority, uses tools, transfers work, records evidence, produces a result, reaches acceptance or dispute, handles exceptions, and closes remediation. If that chain cannot be reconstructed, the system may be observable but not audit-ready.

This paper calls the proposed review unit the **Agentic Audit Object**. It is a structured representation of lifecycle-responsibility-linked agent work. It is not a mandatory data schema. It is a conceptual object model that helps audit, assurance, governance, and technology teams ask whether the relevant lifecycle work can be reviewed.

The Agentic Audit Object exists because agentic systems blur the boundary between process and output. A model-generated draft may be an output. A tool action that sends the draft, records the transaction, updates a system, triggers a workflow, or changes a downstream state is lifecycle work. The auditability question changes from "Was the output acceptable?" to "Can the lifecycle work that produced and accepted that outcome be reconstructed?"

The object shift also changes the meaning of responsibility. Responsibility cannot be inferred from a username, service account, model name, trace ID, or tool log alone. A responsibility-linked review object should identify the human role responsible for intent, authority, review, acceptance, exception handling, and remediation. It should separately identify the agent role and tool surface that executed the work. It should preserve the difference between technical

execution and business responsibility.

Provenance concepts help explain why this structure matters. Provenance language distinguishes entities, activities, agents, and relationships. Audit evidence language asks whether evidence is relevant and reliable for the review objective. this paper combines those ideas with GAIC's MRO layer: lifecycle work must be addressable, linked, partitioned, and reviewable.

The Agentic Audit Object therefore contains more than a log. It contains the work unit, the authority boundary, the human role, the agent role, the tool-action record, the evidence pointer, the accepted outcome state, the exception state, the remediation closure state, and the privacy/selective disclosure profile. Those fields may live in different systems. The model is not saying every enterprise must store them in one database. It says that the relationships must be reconstructable.

This object shift is also where this paper becomes distinct from the Global AI Compliance White Paper 2026. The Global AI Compliance White Paper 2026 identifies the missing regulatory objects for agentic lifecycle governance. This paper asks how those objects become evidence. The same MRO can be read as a governance gap, an engineering object, or an audit evidence object. This paper focuses on the third reading.

Agentic Audit Object Overview

Object component	Review question	Evidence examples	Source / synthesis status	Boundary
Lifecycle work unit	What work is under review?	Work unit ID, scope, task intent, lifecycle stage	author synthesis in this paper grounded by EVID-01 and GAIC-SOURCE	Not mandatory schema
Authority boundary	Was the action authorized within scope?	Delegation record, condition, expiry, escalation path	GAIC-derived + AI-01/AI-08	Not proof of legal delegation
Human role responsibility	Who owned intent, review, acceptance, record,	Role map, review record,	GAIC-derived + AUD-06/AUD-07	Not legal liability

	or closure?	acceptance owner		assignment
Agent role responsibility surface	What agent role executed or transformed the work?	Agent role ID, constraints, capability boundary	GAIC-derived author synthesis + EVID-01	Agent is not a legal person
Tool-action evidence	What external or consequential action occurred?	Tool call record, affected system, reversibility, rollback evidence	GAIC-derived + EVID-02/EVID-03	Not legal liability conclusion
Accepted outcome	Was the result accepted, disputed, rejected, or remediated?	Acceptance state, reviewer role, dispute reason	GAIC-derived + AUD-01/AUD-04	Not compliance proof
Exception / remediation closure	How was deviation handled and closed?	Exception record, corrective action, recheck, closure owner	GAIC-derived + EVID-04/AUD-04	Not legal settlement

Table note: Mixed GAIC-derived and author-synthesis table grounded in audit evidence, provenance, and governance sources. It is not an audit standard or legal template.

Cross-Links

- Chapter 5 expands the object model fields.
- Chapter 6 maps each MRO into audit evidence objects.
- Appendix B provides a walkthrough template.

Boundary Note

The Agentic Audit Object is a proposed object model in this paper for reviewability. It does not create legal categories, professional requirements, certification criteria, or mandatory implementation schema.

4. Why Logs Are Not Audit Evidence Chains

Candidate source status: R4 internal candidate source

Chapter purpose: Ground the core this paper claim that logs and traces are useful but insufficient for agentic auditability.

Reader question: What can logs show, and what do they fail to prove?

Key Claims

- Logs, traces, metrics, and workflow histories can support auditability.
- They do not automatically prove authority, responsibility, accepted outcome, exception handling, privacy treatment, or remediation closure.
- Audit Evidence Chains increase evidentiary value by linking activity to lifecycle responsibility metadata.

Source Grounding Note

Audit evidence language is grounded in AUD-01 and AUD-02. Observability and logging context is grounded in EVID-02 and EVID-03. Provenance concepts are grounded in EVID-01.

Incident/governance context is grounded in EVID-04. AI Act logging context may be referenced as legal context only through AI-06.

Author Synthesis Note

"Logs are not audit evidence chains" is author synthesis in this paper. It is grounded in the distinction between observability/logging and professional audit evidence concepts.

Candidate Prose

Logs are necessary but not sufficient. That sentence is the safest starting point for agentic auditability. The point is not to diminish logs, traces, metrics, monitoring systems, workflow histories, or observability platforms. They are often the first evidence ingredients available. Without them, reconstruction may collapse into narrative memory.

The problem is that logs usually describe activity, not responsibility. A log may show that a service called a tool at a time. A trace may show that a workflow crossed several components. A metric may show latency, error rate, or throughput. An incident record may show that something failed. Those are useful signals. They do not automatically tell a reviewer whether the action was authorized, which human role owned it, which agent role was acting, what business scope applied, whether the tool action was reversible, whether the outcome was

accepted, what privacy treatment applied, or whether remediation closed.

Audit evidence is tied to a review objective. In professional audit language, evidence is evaluated for relevance and reliability within context. For agentic systems, the review objective is not merely to explain system behavior. It is to reconstruct lifecycle responsibility. That means the evidence must connect the technical event to authority, role, tool action, outcome state, exception state, privacy treatment, and closure.

Observability explains how systems behave. Auditability asks whether lifecycle work can be reviewed. The same trace can be valuable for both, but it plays different roles. In observability, the trace helps engineers understand execution path and performance. In auditability, the trace is an input to a larger evidence chain. It must be joined with responsibility records, authority records, evidence pointers, acceptance records, and exception/remediation records.

Consider a tool call. A log may show the tool name, timestamp, endpoint, response code, and service identity. The auditability question is broader: Who authorized the tool action? Was the agent operating inside delegated scope? Which data or system was affected? Was the action reversible? Did a human review occur before or after the action? Was the result accepted, disputed, or remediated? Was sensitive data minimized or redacted in review evidence? Was there a closure state?

The same distinction applies to incident records. An incident log may show that an error occurred. It may not show whether the affected outcome was withdrawn, corrected, re-reviewed, accepted, escalated, or closed. It may not show whether the issue was due to authority drift, responsibility transfer, cross-project reuse, tool-action side effects, privacy leakage, or vendor/runtime substitution. Without linkage to lifecycle objects, the incident is visible but not fully reconstructable.

This paper calls the needed structure an **Audit Evidence Chain**. An Audit Evidence Chain is not just a sequence of logs. It is a responsibility-linked reconstruction path. It ties a lifecycle work unit to the relevant evidence objects: authority boundary, human role, agent role, tool action, evidence pointer, accepted outcome, exception, remediation, privacy treatment, and third-party review boundary where relevant.

This structure also protects against over-retention. A naive response to auditability gaps is to collect everything. That can create privacy and confidentiality risks. A better approach is to preserve evidence pointers, partition keys, disclosure profiles, redaction rules, retention logic,

and integrity records so reviewers can reconstruct what they are authorized to inspect without unnecessary exposure.

Table 2: Logs vs Audit Evidence Chains

Evidence item	What logs/traces can show	What they do not automatically prove	Required linkage	Related MRO	Source / synthesis note
Tool-call log	Tool, endpoint, timestamp, service, response	Business authority, human owner, reversibility, accepted outcome	Delegated authority record, human role map, tool-action evidence, outcome state	MRO-02, MRO-05, MRO-04	EVID-02/EVID-03 + GAIC-SOURCE
Trace span	Execution path, service dependencies, timing	Lifecycle responsibility or review sufficiency	Work unit ID, agent role, evidence pointer, role map	MRO-01, MRO-03, MRO-08	EVID-01/EVID-02 + author synthesis in this paper
Workflow state	Step completion, state transition	Whether state is a governance acceptance state	Acceptance criteria, reviewer role, dispute/remediation state	MRO-04, MRO-16	AUD-01/AUD-04 + GAIC-SOURCE
Incident log	Error event, detection, response action	Remediation closure or legal fault	Exception owner, corrective action, recheck, closure acceptance	MRO-07, MRO-16	EVID-04/AUD-04
Access	User/service	Delegated	Authority	MRO-02,	AI-01/AI-08 + author

log	access event	business authority or scope	boundary,	MRO-07	synthesis in this paper
			confirmation rule, risk class, expiry		
Retention log	Data/evidence storage or deletion event	Whether retention/disclosure is legally sufficient	Privacy treatment, minimization rule, retention basis, disclosure profile	MRO-10, MRO-12, MRO-13	PRIV-01 to PRIV-05

Table note: Mixed source-grounded and author-synthesis table. Logs are treated as useful inputs, not as useless artifacts. The table is not a vendor critique or audit procedure.

Cross-Links

- Appendix A lists evidence request categories.
- Chapter 9 addresses evidence partitioning.
- Chapter 10 addresses minimization and selective disclosure.

Boundary Note

This chapter does not claim observability vendors fail, logs are useless, or telemetry can never support audit evidence. It claims only that raw logs do not automatically become responsibility-linked audit evidence chains.

5. The Agentic Audit Object Model

Candidate source status: R4 internal candidate source

Chapter purpose: Define the proposed object model for audit-ready agentic lifecycle work.

Reader question: What makes lifecycle work audit-ready?

Key Claims

- Audit-ready agentic lifecycle work should be addressable, reconstructable, partitioned,

privacy-aware, and reviewable.

- The Agentic Audit Object Model binds lifecycle work unit, authority boundary, human role responsibility, agent role responsibility, tool-action evidence, evidence pointer, accepted outcome, exception state, remediation closure, and privacy treatment.
- The model is a proposed review architecture, not a mandatory schema or standard.

Source Grounding Note

The model uses audit evidence language from AUD-01, control vocabulary from AUD-04, AI governance context from AI-01, and provenance structure from EVID-01. MRO dependencies are GAIC-derived.

Author Synthesis Note

The Agentic Audit Object Model is author synthesis in this paper. It is a conceptual object model for auditability, not a required implementation format.

Candidate Prose

The Agentic Audit Object Model translates lifecycle work into a reviewable structure. It does not require that every enterprise use the same database table, evidence pack format, or protocol. It asks whether the relationships needed for review can be reconstructed.

The first requirement is an addressable lifecycle work unit. A reviewer needs to know what work is under review: the task, workflow, agent run, delegated action, tool call, handoff, or remediation event. Without a stable work unit, evidence remains scattered across logs, tickets, approvals, messages, policies, and memories.

The second requirement is an authority boundary. Agentic systems often combine technical permission with business action. Technical permission answers whether a service can call a tool. Business authority answers whether that action was allowed under scope, condition, risk class, and responsibility. Auditability requires evidence of the authority boundary, not merely evidence that the tool call succeeded.

The third requirement is role mapping. Human role responsibility and agent role responsibility surface must be separated. The human role owns intent, approval, review, acceptance, escalation, or closure. The agent role describes bounded execution capability, constraints, tool access, and evidence obligations. Confusing these roles weakens accountability and makes review harder.

The fourth requirement is tool-action evidence. A tool action is often where AI-generated output becomes operational consequence. The evidence object should identify the tool, action, target system, affected data or process, initiator, authority basis, reversibility, rollback path, and owner. This does not assign legal liability. It creates reviewability.

The fifth requirement is an evidence pointer. Auditability does not always require raw evidence to be duplicated into a central repository. It may require pointers to evidence, integrity hashes, redaction profiles, retention rules, and partition keys. The pointer must be strong enough that authorized reviewers can reconstruct the chain.

The sixth requirement is an accepted outcome state. An output is not the same as an accepted outcome. The model should distinguish produced, reviewed, accepted, rejected, disputed, remediated, and closed states. This is especially important when a downstream process treats agent work as completed before a responsible human or governance workflow has accepted it.

The seventh requirement is exception and remediation closure. Auditability is weak if exceptions disappear into operational tickets without lifecycle linkage. Exceptions should connect to authority, role, affected outcome, corrective action, recheck, closure owner, and reopen criteria.

The eighth requirement is privacy treatment. Evidence is not improved by unlimited collection. Reviewability must be balanced with minimization, selective disclosure, redaction, retention, and access control. The object model should identify which evidence can be disclosed, to whom, for what review purpose, and under what boundary.

Taken together, these fields create the Audit Evidence Chain. The chain does not replace professional judgment about sufficiency. It makes judgment possible by giving reviewers a reconstructable path.

Agentic Audit Object Model Table

Field	Purpose	Example evidence	Related MRO	Source / synthesis status	Boundary
Lifecycle work unit ID	Names the work under review	Workflow/run/task ID, scope, intent	MRO-08	author synthesis in this paper grounded in EVID-01	Not mandatory schema

Authority boundary	Records delegated scope and limits	Delegation record, condition, expiry, escalation rule	MRO-02, MRO-07	GAIC-derived + AI-01/AI-08	Not legal delegation proof
Human role responsibility	Maps human ownership	Intent owner, review owner, acceptance owner, closure owner	MRO-01	GAIC-derived + AUD-06/AUD-07	Not legal liability assignment
Agent role responsibility surface	Defines bounded agent function	Agent role, constraints, capability boundary, evidence duties	MRO-03, MRO-06	GAIC-derived author synthesis + EVID-01	Agent is not human role
Tool-action evidence	Records consequential action	Tool ID, target system, affected data, rollback path	MRO-05	GAIC-derived + EVID-02/EVID-03	Not liability conclusion
Evidence pointer	Links to review evidence	URI, hash, evidence pack reference, partition key	MRO-08, MRO-12	Mixed	Not blanket disclosure
Accepted outcome	Records outcome governance state	Accepted/ rejected/ disputed/ remediated status	MRO-04	GAIC-derived + AUD-01/AUD-04	Not compliance proof

Exception state	Records deviation or dispute	Exception ID, trigger, owner, escalation path	MRO-07, MRO-16	GAIC-derived + EVID-04	Not legal breach finding
Remediation closure	Records corrective action and closure	Corrective action, recheck, closure reviewer, reopen criteria	MRO-16	GAIC-derived + AUD-04/AUD-05	Not settlement or regulator closure
Privacy treatment	Defines minimization and disclosure	Redaction profile, retention rule, access scope	MRO-10 to MRO-13	PRIV sources + GAIC synthesis	Not legal advice

Table note: This table is mixed source-grounded, GAIC-derived, and author-synthesis. It is a drafting model, not an audit standard, certification checklist, or legal template.

Cross-Links

- Chapter 6 maps the model to all MROs.
- Appendix A converts fields into evidence requests.
- Appendix B turns fields into a walkthrough template.

Boundary Note

The Agentic Audit Object Model should be read as a review architecture. It does not require MPLP, Cognitive OS, Validation Lab, or any particular vendor system.

6. MRO-to-Audit-Evidence Mapping

Candidate source status: R4 internal candidate source

Chapter purpose: Translate GAIC MROs into audit evidence objects and evidence requests.

Reader question: How does each MRO become evidence?

Key Claims

- GAIC MROs can be mapped into reviewable audit evidence objects.

- The mapping turns lifecycle governance objects into evidence request categories.
- MROs remain GAIC-derived governance objects; they are not legal mandates, audit standards, or certification criteria.

Source Grounding Note

MRO names and meanings use GAIC source truth. Audit evidence language uses AUD-01. AI audit/control context uses AUD-05 and AI-01.

Author Synthesis Note

The MRO-to-audit-evidence mapping is author synthesis in this paper derived from GAIC objects and externally sourced audit/control terminology.

Candidate Prose

The Global AI Compliance White Paper 2026 introduced the sixteen Missing Regulatory Objects for agentic and multi-agent system governance. This paper reads those objects through an auditability lens. Each MRO asks a governance question. This paper adds the evidence question: what object would a reviewer need to inspect to reconstruct the lifecycle work?

This translation is not a claim that MROs are law. It is not a claim that the presence of an MRO proves compliance. It is not a claim that absence proves non-compliance. MROs are GAIC-derived lifecycle responsibility objects. Their value for this paper is that they give auditability a vocabulary for the work that logs alone do not capture.

The mapping also prevents a common failure in agentic governance: treating "evidence" as a flat export. Agentic systems do not need a pile of logs; they need evidence structured around review questions. Who owned the action? What authority applied? Which agent role acted? What tool created consequence? Was the outcome accepted? What exception occurred? What data was exposed? What changed after vendor, model, or runtime substitution? How did remediation close?

The sixteen MROs can be grouped into six auditability clusters:

1. Responsibility and role mapping: MRO-01, MRO-03, MRO-06.
2. Authority and action boundaries: MRO-02, MRO-05, MRO-07.
3. Outcome and closure: MRO-04, MRO-16.
4. Evidence partitioning and reuse: MRO-08, MRO-09.

- 5. Privacy and disclosure: MRO-10, MRO-11, MRO-12, MRO-13.
- 6. Vendor, processor, and substitution chains: MRO-14, MRO-15.

This cluster view helps readers see that auditability is not one control. It is the ability to reconstruct a lifecycle. A work unit can fail auditability at many points: no role map, no authority record, no accepted outcome state, no evidence partition, no privacy treatment, no substitution record, or no remediation closure.

Table 3: MRO-to-Audit-Evidence Mapping

MRO	Audit evidence object	Sample evidence request	Source support	Guide 1 relation	Guide 2 relation	Bo
MRO-01 Human Role to MAS Responsibility Mapping	Human-role responsibility map	Show accountable role for intent, authorization, review, acceptance, exception, and remediation	AUD-06, AUD-07, AI-01	Role/evidence architecture	Responsibility governance	No ass
MRO-02 Delegated Authority Boundary	Delegated authority record	Show scope, condition, expiry, escalation, and reauthorization evidence	AI-01, AI-06, AI-08, AUD-06	Authority control implementation	Delegation policy-to-evidence	Not per
MRO-03 Agent Role is not Human Role	Agent role boundary object	Show agent role, capability limits, constraints, owner, and escalation	AI-08, EVID-01, AI-01	Agent role design	Accountability model	Do ant age

		boundary				
		Show				
		acceptance				
MRO-04 Accepted Outcome Compliance	Accepted outcome record	criteria, reviewer, acceptance/rejection/dispute state, and evidence link	AUD-01, AUD-02, AUD-04	Outcome state capture	Review/ acceptance governance	Acco com
		Show tool				
		identity, action, affected	EVID-02			
MRO-05 Tool-Action Liability Boundary	Tool-action evidence object	system/data, authority basis, reversibility, rollback path, owner	EVID-03, EVID-04, AUD-05	Tool evidence capture	Tool-action approval policy	No com
		Show				
		source/target agent, transferred				
MRO-06 Responsibility Transfer Across Agents	Responsibility transfer record	scope, retained scope, inherited constraints, acceptance/rejection	EVID-01, AI-08, AUD-06	Handoff record design	Transfer governance	Not
		Show baseline				
MRO-07 Authority Drift	Authority drift exception record	authority, observed action, trigger, ,	AI-01, AI-02, EVID-04	Drift monitoring	Exception governance	Not leg

		escalation, stop/downgrade/reauthorization	AUD-04			
		Show partition keys, evidence pointers, link identifiers, integrity hashes, access rules	EVID-01 EVID-02 EVID-03 AUD-01	Evidence partitioning architecture	Evidence access governance	No rete e
MRO-08 MAS Evidence Partitioning	Partitioned evidence chain					
		Show source context, target context, reset validation, privacy review, authorization refresh	AI-01, AI-05, PRIV-02 AUD-04	Reuse validation workflow	Reuse governance	No clea
MRO-09 Cross-Project Reuse Compliance	Reuse context validation record					
		Show data flow, processing purpose, retention rule, rights workflow, minimization treatment	PRIV-01 to PRIV-05 , AI-09	Privacy-aware evidence capture	Privacy governance	No GD
MRO-10 Privacy / GDPR Lifecycle Mapping	Privacy lifecycle evidence object					
		Show ruleset identity, redaction profile, evidence	PRIV-02 , PRIV-03 , BOUND	Evidence export/replay	Third-party review governance	No clai
MRO-11 Privacy-Preserving Third-Party Validation	Validation disclosure profile and verdict record					

		pointer, verdict hash, reviewer boundary	-01, BOUND -02, AUD-03			
MRO-12 Evidence Minimization and Selective Disclosure	Evidence minimization and disclosure object	Show minimization rule, disclosure justification, redaction profile, access log, expiry	PRIV-01 to PRIV-05	Selective disclosure design	Disclosure policy	No con
MRO-13 Data Subject Rights vs Evidence Retention	Rights-retention reconciliation record	Show request type, retention basis, hold status, deletion/redacti on action, owner	PRIV-01 , PRIV-02 , PRIV-05 , AI-09	Rights-aware retention	Legal/privacy governance	Re rev
MRO-14 Third- Party Processor / Subprocessor Chain	Processor/ subprocessor chain evidence	Show processor role, subprocessor chain, data agreement pointer, responsibility owner	AI-09, PRIV-01 , PRIV-02 , AUD-04	Vendor evidence partitioning	Processor governance	No
MRO-15 Vendor / Model / Runtime Substitution Conformance	Substitution conformance record	Show prior/new component, control evidence, evidence-	AI-01, AI-04, AI-05, AUD-04, BOUND	Runtime/model change evidence	Change governance	No cer

			integrity check,			
			reauthorization -02			
			, revalidation			
			Show event,			
			affected			
			outcome,			
			owner,	EVID-04		
MRO-16 Incident,	Incident/		corrective	, AI-02,	Remediation	No
Dispute, and	remediation		action,	AUD-04,	workflow	reg
Remediation	closure record	recheck,		AUD-05,	evidence	governance
Closure		closure		AUD-06		acc
		acceptance,				
		reopen				
		condition				

Table note: GAIC-derived and author-synthesis mapping grounded in audit/control/governance source language. It is not a legal checklist, audit procedure, certification criterion, or procurement tool.

Cross-Links

- Appendix C contains the expanded mapping.
- Chapter 13 maps the same objects into AARM dimensions.
- Guide 1 will later translate this table into implementation planning.
- Guide 2 will later translate it into compliance operating-model planning.

Boundary Note

This chapter does not invent new MRO numbering. It uses current GAIC source truth for MRO-01 through MRO-16.

7. Evidence Request List for Agentic Systems

Candidate source status: R4 internal candidate source

Chapter purpose: Define the logic of evidence requests for agentic systems without turning

them into formal audit procedures.

Reader question: What should reviewers ask to reconstruct agentic lifecycle work?

Key Claims

- Evidence requests should be organized by lifecycle stage and review objective.
- Agentic evidence requests must cover role, authority, agent/tool action, outcomes, exceptions, privacy, partitioning, and closure.
- Evidence requests are readiness architecture, not universal legal demands or formal audit procedures.

Source Grounding Note

Audit evidence and evidence sufficiency language is grounded in AUD-01 and AUD-02. Control and AI audit practice context comes from AUD-04, AUD-05, and AUD-06. AI governance/logging context may use AI-06.

Author Synthesis Note

The evidence request taxonomy is author synthesis in this paper derived from GAIC MROs and professional audit/control language.

Candidate Prose

Agentic auditability becomes practical when it is expressed as evidence requests. A high-level statement that "the system is logged" is not enough. A reviewer needs to know what evidence should exist, where it lives, which lifecycle stage it supports, what role owns it, how sensitive fields are treated, and what boundary applies.

The evidence request list should begin with the lifecycle work unit. The reviewer should be able to ask: What work was initiated? What purpose and scope did it have? Who initiated it? Which agent or workflow received it? Which policy, authority, or business context applied? Without a work unit, later evidence cannot be connected.

The second request category is authority. What delegated authority allowed the agent to act? Was the action within scope? Did the authority have conditions, risk class, expiry, confirmation, or escalation requirements? If authority changed, was the change recorded? If authority drift occurred, was it escalated and closed?

The third category is responsibility. Which human role owned intent, review, acceptance,

exception handling, and remediation? Which agent role executed the work? Was a responsibility transfer recorded across agents? Was a tool action tied back to a human or organizational owner? This category distinguishes accountability from technical execution.

The fourth category is tool-action evidence. Tool calls are where agent behavior often becomes operational consequence. Evidence should identify the tool, action, target system, affected data or process, response, reversibility, rollback path, and owner. The request is not asking for legal liability conclusions. It is asking for reconstructability.

The fifth category is outcome evidence. Reviewers should distinguish produced output from accepted outcome. Was the output accepted, rejected, disputed, escalated, or remediated? Who reviewed it? Which criteria or policy were applied? What evidence supports the final state?

The sixth category is exception and remediation. What event occurred? What lifecycle object was affected? Who owned the exception? What corrective action was taken? Was the action rechecked? Who accepted closure? What would reopen the issue?

The seventh category is privacy and selective disclosure. Evidence should not be over-collected or over-disclosed. Reviewers should ask for data categories, redaction profiles, retention rules, disclosure scope, access logs, and minimization rationale. Legal interpretation remains outside this paper, but the evidence architecture should make privacy-aware review possible.

Finally, evidence requests should include boundary statements. Each request should clarify whether it supports readiness, reconstruction, control review, third-party review, or assurance planning. It should also clarify what it does not prove.

Table 4: Lifecycle Stage -> Audit Evidence Request

Lifecycle stage	Core review question	Evidence requested	Human role	Agent/tool record	Outcome/ closure	Privacy treatment
Intent	What work was initiated and why?	Work unit ID, purpose, scope, initiating context	Request owner	Initial agent/workflow assignment	No accepted outcome yet	Purpose and data categories
Delegation	Was action	Authority	Delegating/	Agent	Escalation	Minimum

	authorized within scope?	grant, conditions, expiry, confirmation boundary	approving role	permission and tool scope	path	necessary authority evidence
Planning	What plan or route was selected?	Plan version, constraints, review flags	Plan reviewer if required	Agent plan trace	Plan accepted/rejected state	Redacted sensitive prompts if needed
Tool action	What external consequence occurred?	Tool identity, action, target, response, affected data, reversibility	Tool-action owner	Tool-call record, service trace	Action status and rollback path	Data minimization and access scope
Handoff	Was responsibility transferred?	Source/target agent, transferred scope, retained scope, acceptance/rejection	Handoff owner	Agent transfer record	Transfer state	Partition and disclosure rule
Review	Was work reviewed before acceptance?	Review record, criteria, evidence pointer	Reviewer role	Supporting traces/outputs	Accepted/rejected/disputed	Redaction profile

Exception	What deviation occurred?	Exception record, baseline, trigger, impact	Escalation owner	Incident trace or alert	Open/ remediated/ closed	Sensitive-data handling
	Closure	Was remediation complete?	Corrective action, recheck evidence, closure owner, reopen criteria	Closure owner	Remediation workflow trace	Closure state

Table note: Mixed source-grounded and author-synthesis table. It is a readiness evidence architecture, not a formal audit procedure, legal demand, or certification criterion.

Cross-Links

- Appendix A expands the evidence request list.
- Appendix B turns these categories into a walkthrough template.
- Appendix E provides exception/remediation checklist detail.

Boundary Note

Evidence requests do not prove legal compliance or audit sufficiency. A qualified professional must evaluate sufficiency in context.

8. Lifecycle Walkthrough for AI Agent / MAS Work

Candidate source status: R4 internal candidate source

Chapter purpose: Provide a reconstruction pattern for one agentic lifecycle run or work unit.

Reader question: How is one agentic workflow walked from intent to accepted outcome?

Key Claims

- A lifecycle walkthrough should connect intent, authority, planning, execution, handoff, evidence, review, outcome, exception, remediation, and closure.
- The walkthrough is illustrative and readiness-oriented, not a formal audit procedure.
- Walkthroughs should preserve human role and agent role separation.

Source Grounding Note

Control and walkthrough-adjacent language uses AUD-04, AUD-05, and AUD-06. Provenance and trace context uses EVID-01 and EVID-02. GAIC source truth provides MRO lifecycle objects.

Author Synthesis Note

The agentic lifecycle walkthrough is author synthesis in this paper. It translates GAIC MROs and evidence request categories into a reconstruction pattern.

Candidate Prose

A lifecycle walkthrough tests whether a single unit of agentic work can be reconstructed. It begins with a concrete work unit: a task, run, agent workflow, multi-agent handoff, tool action, or remediation event. The goal is not to prove compliance. The goal is to determine whether the evidence chain can explain the work from intent to closure.

The walkthrough should begin with intent and scope. What was the system asked to do? Who initiated the work? What business process, policy, or governance context applied? What was out of scope? The purpose of this step is to avoid reviewing an isolated output without understanding the work it belongs to.

The second step is authority. What authority allowed the agent or workflow to act? Was authority delegated by role, policy, system configuration, human confirmation, or another control? Did that authority include limits, expiry, risk class, confirmation requirements, or escalation paths? If the tool action exceeded scope, was that captured as exception or authority drift?

The third step is agent and tool identification. Which agent role acted? What constraints applied to that role? Which tools were available? Which tools were used? Which external systems, data stores, APIs, or processors were touched? The walkthrough should separate the agent role from the human role and the tool surface.

The fourth step is evidence pointer review. Where are logs, traces, workflow records, approvals, policies, tickets, evidence packs, or hash manifests stored? Are they linked by a stable work unit ID? Are there partition keys by agent, tool, role, vendor, project, data class, and lifecycle stage? Can an authorized reviewer reconstruct the chain without overexposing sensitive data?

The fifth step is outcome review. What output or action occurred? Was the outcome accepted, rejected, disputed, escalated, remediated, or closed? Who had authority to accept it? What evidence supports that state? An accepted outcome is a governance state, not just a completed workflow event.

The sixth step is exception and remediation review. If something deviated, did the system record the baseline, trigger, owner, corrective action, recheck, closure state, and reopen criteria? A lifecycle walkthrough is incomplete if it stops at detection and never reaches closure.

The seventh step is boundary review. What does the walkthrough support? It may support readiness discussion, internal review, evidence request scoping, or future assurance planning. It does not itself produce an audit opinion, certification, legal conclusion, or regulator-approved result.

Table 5: Human Role / Agent / Tool Responsibility Matrix

Lifecycle task	Human role	Agent role	Tool/system	Authority basis	Evidence pointer	Escalation path
Initiate work	Request owner	Intake agent	Workflow system	Business request policy	Work unit record	Reject or request clarification
Approve external action	Business owner or delegated approver	Execution agent	CRM/API/ configuration tool	Delegated authority record	Authority evidence pointer and tool log	Human confirmation or stop
Transfer task	Process owner	Source and target agents	Orchestration runtime	Handoff rule and inherited constraints	Responsibility transfer record	Return/ reject/ escalate

Review output	Reviewer role	Drafting or analysis agent	Document/ review system	Review policy	Output, review note, evidence pointer	Dispute or remediation
Handle exception	Incident/ remediation owner	Monitoring or remediation agent	Ticketing/ incident system	Escalation rule	Exception record and remediation evidence	Reopen, escalate, or close

Table note: GAIC-derived and author-synthesis table grounded in internal audit, control, provenance, and observability context. It assigns review responsibility for auditability purposes only and does not assign legal liability.

Walkthrough Output

A completed walkthrough should produce:

- Work unit identity and scope.
- Authority boundary and confirmation history.
- Human role and agent role map.
- Tool-action evidence and side-effect summary.
- Evidence chain with pointers and partition keys.
- Accepted outcome or dispute state.
- Exception/remediation closure state if applicable.
- Privacy treatment and disclosure profile.
- Boundary statement describing what the walkthrough does and does not prove.

Cross-Links

- Appendix B provides the walkthrough template.
- Chapter 9 addresses partitioning across agents, tools, vendors, and projects.
- Chapter 11 addresses exceptions and remediation closure.

Boundary Note

The walkthrough is a reconstruction pattern for readiness and evidence design. It is not a formal audit procedure, assurance engagement step, or legal compliance test.

9. Evidence Partitioning Across Agents, Tools, Roles, Vendors, and Projects

Candidate source status: R4 internal candidate source

Chapter purpose: Define how evidence is separated and linked across lifecycle boundaries.

Reader question: How do reviewers avoid flattened responsibility in multi-agent and tool-mediated work?

Key Claims

- Evidence must be partitioned and linkable across agent, tool, role, vendor, project, data class, jurisdiction, and lifecycle state where relevant.
- Partitioning enables reconstruction without blanket disclosure.
- Cross-project reuse, third-party processor chains, and vendor/model/runtime substitution require explicit evidence boundaries.

Source Grounding Note

Provenance and observability/logging context uses EVID-01, EVID-02, and EVID-03. Privacy context uses PRIV-01, PRIV-02, PRIV-03, and AI-09. GAIC source truth provides MRO-08, MRO-09, MRO-10, MRO-14, and MRO-15.

Author Synthesis Note

The evidence partitioning model is author synthesis in this paper grounded in provenance, observability, privacy, and GAIC MROs.

Candidate Prose

Agentic evidence becomes difficult to review when everything is flattened into one trace, one log export, one transcript, or one vendor report. Multi-agent systems distribute work across agents, tools, human roles, vendors, processors, projects, and data boundaries. Auditability requires those boundaries to be visible.

Partitioning is the discipline of separating evidence by review-relevant boundary while preserving the links needed for reconstruction. It is not the same as hiding evidence. It is not the same as retaining everything. It is a way to make evidence reviewable, scoped, and privacy-aware.

The first partition is the work unit. Every evidence item should connect to a lifecycle work unit or

reconstruction path. Without a work unit, reviewers may know that events happened but not which lifecycle task they belong to.

The second partition is the agent. Agent roles may differ in authority, capability, tool access, evidence duties, and escalation requirements. If evidence only shows that "the system" acted, responsibility surfaces become blurred. Reviewers should be able to distinguish planning agents, execution agents, review agents, monitoring agents, and remediation agents where those roles exist.

The third partition is the tool. Tool actions often create external consequences. Evidence should separate model output from tool execution, tool execution from downstream system state, and reversible actions from irreversible or high-impact actions. Tool-specific evidence can include request/response logs, target system records, affected data categories, rollback records, and owner information.

The fourth partition is the human role. A user account is not enough. Auditability requires role responsibility: request owner, delegating role, reviewer, acceptor, exception owner, remediation owner, privacy owner, or internal audit reviewer. These roles should connect to evidence without implying legal liability.

The fifth partition is vendor, processor, and runtime boundary. Agentic systems may call third-party tools, external APIs, model providers, orchestration runtimes, vector stores, workflow systems, or processors/subprocessors. Evidence should identify where responsibility, data processing, and review access shift. This is not vendor ranking. It is boundary mapping.

The sixth partition is project and reuse context. Cross-project reuse can move prompts, agents, workflows, memory, tools, or policies into a new context. Reuse may be efficient, but auditability requires evidence that scope, authority, privacy treatment, and evidence obligations were reset or revalidated for the new context.

The seventh partition is privacy and disclosure class. Some evidence can be disclosed directly. Some should be redacted. Some may require evidence pointers, hashes, summaries, or reviewer-specific access. The partitioning model should make it possible to review what is necessary without exposing more than the review requires.

Table 6: Evidence Partitioning Matrix

Partition	Why it matters	Evidence	Access/	Failure if	Related
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dimension		boundary	disclosure rule	missing	MRO
Work unit	Anchors reconstruction	Work unit ID and lifecycle stage	Reviewer access tied to scope	Evidence cannot be connected	MRO-08
Agent	Separates responsibility surfaces	Agent ID, role, constraints, work segment	Role-based evidence disclosure	Flattened agent responsibility	MRO-03, MRO-06, MRO-08
Tool	Tracks external consequence	Tool ID, target system, action type, reversibility	Tool-specific disclosure and redaction	Unclear side effects	MRO-05, MRO-08
Human role	Connects governance ownership	Role map, approval, review, acceptance, closure	Need-to-know reviewer access	User identity mistaken for responsibility	MRO-01
Vendor / processor	Maps third-party boundary	Vendor/processor role, subprocessor chain, data path	Contract/privacy-aware disclosure	Responsibility chain disappears	MRO-14
Project / reuse context	Prevents context drift	Source context, target context, reset validation	Reuse-specific review access	Old authority/policy reused incorrectly	MRO-09
Runtime/ model substitution	Preserves evidence continuity after change	Prior/new component, revalidation evidence	Change-scoped disclosure	Evidence chain breaks after substitution	MRO-15

Privacy class	Limits	Data category,		Privacy leakage or evidence hoarding	MRO-10, MRO-12, MRO-13
	overexposure	redaction	Minimized		
		profile, retention rule	disclosure		

Table note: Mixed source-grounded and author-synthesis table. It does not create universal disclosure or retention rules, vendor rankings, or procurement recommendations.

Cross-Links

- Chapter 10 covers privacy and selective disclosure.
- Appendix A includes partition fields for evidence requests.
- Appendix C maps partitioning to MRO-08, MRO-09, MRO-14, and MRO-15.

Boundary Note

Partitioning is a review architecture. It does not decide legal retention, data subject rights, processor obligations, or vendor liability.

10. Privacy, Selective Disclosure, and Audit Evidence

Minimization

Candidate source status: R4 internal candidate source

Chapter purpose: Balance auditability with privacy, confidentiality, minimization, and retention discipline.

Reader question: How can evidence be reviewable without overexposure?

Key Claims

- Auditability should not become evidence hoarding.
- Privacy-aware auditability uses minimization, redaction, retention discipline, evidence pointers, partitioning, and selective disclosure.
- this paper frames privacy/evidence tension; it does not provide legal advice or GDPR compliance proof.

Source Grounding Note

Privacy and minimization context uses PRIV-01, PRIV-02, PRIV-03, PRIV-04, PRIV-05, and

AI-09. GAIC source truth provides MRO-10 through MRO-13 and the Validation Lab selective disclosure boundary.

Author Synthesis Note

The privacy-preserving audit evidence controls are author synthesis in this paper grounded in public privacy guidance and GAIC MROs.

Candidate Prose

Auditability can create a privacy problem if it is misunderstood. The answer to weak evidence is not to collect everything forever. Agentic systems may process personal data, confidential business data, privileged material, regulated data, customer records, employee records, prompts, outputs, tool responses, logs, memory, and third-party processor data. If auditability becomes indiscriminate evidence capture, it may increase exposure and reduce trust.

The better question is: what evidence is necessary for the review objective, and how can it be disclosed safely? This requires minimization, partitioning, retention discipline, access control, redaction, evidence pointers, and reviewer-specific disclosure profiles.

Privacy-aware auditability begins with purpose. Evidence should be collected and retained for defined review purposes: reconstruction, control review, exception handling, third-party validation, remediation, or assurance planning. A generic desire to "keep all logs just in case" is not a mature evidence strategy.

The second principle is minimization. Evidence objects should identify what fields are needed for reconstruction and which fields can be redacted, tokenized, summarized, hashed, or replaced with pointers. A reviewer may need to know that a tool action affected a customer data category without seeing every raw value. Another reviewer may need access under a narrower approved scope. The evidence architecture should support both.

The third principle is selective disclosure. Different reviewers need different levels of evidence. Internal engineering may need logs and traces. Internal audit may need work unit chains and evidence pointers. Privacy teams may need data category and retention records. Third-party reviewers may need redacted evidence packs, ruleset identity, verdict hashes, or replay packages. The system should make disclosure explicit, scoped, and recorded.

The fourth principle is retention discipline. Evidence needs may conflict with storage limitation and deletion expectations. This paper does not decide legal retention periods. It asks that

systems record retention rules, expiry, holds, deletion/redaction actions, and rights-retention reconciliation where relevant. A reviewable retention decision is stronger than a silent default.

The fifth principle is boundary clarity. Privacy sources provide context, not legal conclusions. A privacy-preserving evidence architecture does not prove GDPR compliance, UK GDPR compliance, EU AI Act compliance, or any jurisdiction-specific legal compliance. It creates the object layer that legal, privacy, audit, and governance professionals can review.

Table 7: Privacy-Preserving Audit Evidence Controls

Control	Purpose	Evidence artifact	Selective disclosure mechanism	Privacy risk reduced	Residual review need
Evidence minimization	Avoid unnecessary collection	Minimization rule, field inventory	Request only review-relevant fields	Overcollection	Legal/privacy review for purpose and necessity
Redaction profile	Remove or mask sensitive fields	Redaction rule and redaction log	Redacted evidence pointer	Overexposure to reviewer	Sufficiency check after redaction
Evidence pointer	Reference evidence without copying raw data	Pointer, hash, manifest entry	Reviewer-scoped access or replay	Data duplication	Integrity and access review
Retention rule	Limit storage duration or define hold	Retention schedule, expiry, hold indicator	Time-bound access	Excessive retention	Legal/privacy retention analysis
Disclosure profile	Define who can see what	Disclosure matrix,	Role-based disclosure	Unscoped sharing	Reviewer authorization

		access log	package		review
	Address				
	tension	Request			
	between	record,	Redaction/	Silent conflict	
Rights-retention	rights	action, hold	deletion/hold	between	Legal review
reconciliation	requests and	status,	decision path	evidence and	
	evidence	rationale		rights	
	retention				
		Redacted			
	Support	evidence			
Third-party	review	pack, ruleset	Non-raw evidence	Third-party	Boundary and
validation	without raw	ID, verdict	adjudication	overexposure	conflict review
package	exposure	hash			

Table note: Source-grounded and GAIC-derived table. It is not legal advice, privacy compliance proof, or universal retention guidance.

Cross-Links

- Chapter 9 addresses evidence partitioning.
- Chapter 12 addresses third-party validation boundaries.
- Appendix F provides privacy/legal boundary language.

Boundary Note

This chapter does not provide legal advice, interpret specific legal obligations, or claim that any architecture proves compliance with GDPR, UK GDPR, EU AI Act, or other privacy laws.

11. Exception, Dispute, and Remediation Closure

Candidate source status: R4 internal candidate source

Chapter purpose: Define closure evidence for failed, disputed, drifted, or remediated agentic work.

Reader question: How does auditability handle exceptions and closure?

Key Claims

- Exceptions, disputes, authority drift, outcome rejection, and remediation must close as evidence-backed lifecycle states.
- Incident logs alone do not prove closure.
- Closure evidence should identify owner, corrective action, recheck, accepted closure state, and reopen criteria.

Source Grounding Note

Incident/governance context uses EVID-04 and AI-02. Control and AI audit practice context uses AUD-04, AUD-05, and AUD-06. GAIC source truth provides MRO-04, MRO-07, and MRO-16.

Author Synthesis Note

Remediation closure as an audit evidence state is GAIC-derived author synthesis. It is not a legal settlement, regulator closure, or assurance conclusion.

Candidate Prose

Agentic auditability is incomplete if it only reconstructs successful work. Many of the most important review questions arise when an agent acts outside expected authority, a tool action creates unintended consequence, an outcome is disputed, a privacy issue appears, a handoff fails, or remediation is required.

In ordinary operations, exceptions often become tickets. Tickets are useful, but they are not automatically lifecycle evidence. A ticket may describe an issue, assign an owner, record comments, and mark a status. For auditability, the ticket must connect to the agentic work unit, authority boundary, role map, tool action, evidence pointer, accepted outcome state, privacy treatment, corrective action, and closure state.

Closure is the critical concept. An incident can be detected without being remediated. A remediation can be attempted without being rechecked. A recheck can occur without being accepted. An accepted closure can later be reopened. Auditability requires those state transitions to be evidence-backed.

For agentic systems, exceptions should be categorized by lifecycle object. Was the issue an authority exception, where observed behavior exceeded delegated scope? Was it a responsibility exception, where no human owner could be identified? Was it a tool-action

exception, where a tool created unexpected side effects? Was it an accepted-outcome dispute, where a result was produced but not accepted? Was it a privacy exception, where evidence collection or disclosure exceeded the intended boundary? Was it a substitution exception, where evidence integrity broke after runtime/model/tool change?

The closure record should answer five questions:

1. What happened and which lifecycle object was affected?
2. Who owned the exception and remediation?
3. What corrective action was taken?
4. What evidence shows recheck or review?
5. Who accepted closure, and what would reopen the issue?

None of these questions determines legal fault. None proves regulatory acceptance. None settles liability. They make the remediation lifecycle reconstructable.

Table 10: Exception / Dispute / Remediation Evidence Checklist

Event type	Evidence required	Owner	Escalation path	Remediation record	Closure evidence	Boundary
Authority exception	Baseline authority, observed action, trigger, affected work unit	Escalation owner	Stop, downgrade, reauthorize, or human confirmation	Reauthorization or stop-action record	Closure acceptance and new boundary	Not legal determination
Outcome dispute	Output, acceptance criteria, dispute reason, reviewer notes	Review owner	Secondary review or remediation	Correction/review path	Accepted, rejected, or remediated state	Not settlement proof
Tool-action	Tool call,	Tool-action	Rollback,	Rollback or	Recheck	Not liability

side effect	target system, affected data/proc ess, reversibilit y	owner	disable, escalate	corrective action	and owner acceptance	finding
Privacy event	Data category, disclosure path, redaction/ retention record	Privacy owner	Privacy/ legal review	Redaction, deletion, hold, or access restriction	Review note and disclosure update	Not legal advice
Handoff failure	Source/ target agent, transfere d scope, retained scope	Process owner	Return, reassign, or escalate	Transfer correction	Accepted transfer or closed exception	Not legal responsibility transfer
Remediation defect	Prior closure evidence, reopened issue, failed recheck	Remediation owner	Reopen and reassign	Updated corrective action	Recheck and closure reviewer	Not regulator closure

Table note: Mixed source-grounded, GAIC-derived, and author-synthesis checklist. It is not an audit procedure, legal remedy checklist, or certification criterion.

Cross-Links

- Appendix E provides an expanded closure checklist.
- Chapter 13 includes exception traceability and remediation closure as AARM dimensions.
- Chapter 16 connects closure to future Guide 1, Guide 2, and the later insurability white paper.

Boundary Note

Closure in this chapter means evidence-backed governance closure. It does not mean legal closure, settlement, regulator acceptance, assurance opinion, or certification.

12. Third-Party Validation Without Certification Claims

Candidate source status: R4 internal candidate source

Chapter purpose: Define how evidence review can involve third parties without becoming certification, assurance opinion, conformity assessment, legal proof, or regulator approval.

Reader question: What can third-party validation support without becoming certification?

Key Claims

- Third-party validation can support evidence review, replay, adjudication, and readiness discussion.
- Third-party validation does not certify compliance or issue assurance opinions unless a separate authorized engagement or conformity-assessment framework exists.
- Validation Lab is only a non-certifying evidence adjudication example.

Source Grounding Note

Assurance and attestation boundaries are grounded in AUD-03 and BOUND-03. Conformity-assessment and validation/verification boundaries are grounded in BOUND-01 and BOUND-02. Validation Lab boundary is GAIC source truth.

Author Synthesis Note

this paper's third-party validation boundary is author synthesis that uses professional boundary sources to avoid overclaim. It does not define a conformity assessment program.

Candidate Prose

Agentic auditability often benefits from independent review. A third party, internal audit team, governance function, or specialized review group may inspect evidence packs, replay work units, test whether evidence objects exist, adjudicate ruleset conformance, or evaluate whether evidence chains are reconstructable. These activities can be valuable.

They are also easy to overstate. Evidence review is not automatically certification. Validation is not automatically an assurance opinion. A verdict hash is not regulator approval. A ruleset check is not legal compliance proof. A third-party review note is not an audit report unless it is performed under an appropriate professional engagement, criteria, scope, independence framework, and reporting structure.

This chapter exists to keep that boundary visible. This paper uses "third-party validation" in a narrow sense: review of evidence against defined rulesets or reconstruction objectives. The review may support readiness, evidence quality, selective disclosure, replay, exception closure, or future assurance planning. It does not itself create certification or assurance.

Validation Lab is a useful boundary example because GAIC already frames it as non-certifying evidence adjudication. In this paper, Validation Lab should only be described as one example of how evidence-based validation might be operationalized. It must not be described as the only validation path, a certification body, a regulator, an audit firm, an assurance provider, a conformity assessment body, or proof of enterprise readiness.

The distinction matters for enterprises. If an organization receives a third-party evidence review, it should know exactly what was reviewed, against what ruleset, under what scope, with what evidence, and with what limitations. A review may show that evidence exists and can be replayed. It may show that a ruleset passed. It may show that disclosure was minimized. But it does not determine legal compliance or professional audit sufficiency.

The same distinction matters for reviewers. A review team should state its role: internal readiness review, independent evidence review, validation, verification, assurance engagement, attestation engagement, certification, or legal analysis. Each role has different requirements. this paper only defines the auditability object layer that may support those roles.

Table 8: Third-Party Validation Boundary Table

Review activity	Can support	Cannot prove	Evidence artifact	Boundary language	Source IDs
Evidence adjudication	Ruleset conformance review	Legal compliance or audit opinion	Ruleset ID, evidence pointer, verdict/hash record	Non-certifying evidence review	GAIC-SOURCE, BOUND-01
Evidence replay	Reconstructability check	Operating effectiveness across all periods	Replay package, work unit chain, integrity record	Scope-limited reconstruction	EVID-01, AUD-01
Selective disclosure review	Privacy-aware review process	Privacy law compliance	Redaction profile, disclosure log, reviewer access scope	Privacy-aware evidence review, not legal advice	PRIV-01 to PRIV-05
Third-party readiness review	Independent input to readiness discussion	Assurance conclusion	Scope statement, findings, limitations	Readiness input only	AUD-03, BOUND-03

Validation Lab example	Non-certifying evidence adjudication example	Certification, regulator approval, conformity assessment, legal proof	Verdict hash, ruleset identity, evidence pointer	One optional example, not exclusive path	GAIC-SOURCE, BOUND-01, BOUND-02
Assurance engagement	Professional assurance under separate criteria and engagement terms	Anything outside engagement scope	Practitioner report, criteria, scope, evidence basis	Outside this paper unless separately authorized	AUD-03, BOUND-03

Table note: Source-grounded boundary table with GAIC-derived example. It is not a certification model, audit procedure, or assurance engagement template.

Cross-Links

- Appendix F provides reusable boundary language.
- Appendix D uses third-party review boundary as an AARM dimension.
- Chapter 15 explains professional use by audit and assurance firms.

Boundary Note

This chapter must not be read as claiming that any third party, including Validation Lab, certifies agentic systems, issues legal opinions, provides assurance opinions, or has regulator approval.

13. Agentic Auditability Readiness Model

Candidate source status: R4 internal candidate source

Chapter purpose: Present AARM dimensions and readiness levels as a proposed readiness model.

Reader question: How can auditability readiness be described without claiming assurance?

Key Claims

- AARM describes readiness for reconstructability, testability, evidence structure, privacy-aware review, and third-party review boundaries.
- AARM is proposed author synthesis in this paper.
- AARM is not a score, audit standard, certification, assurance opinion, legal compliance proof, regulator approval, procurement tool, or vendor ranking.

Source Grounding Note

Audit evidence language uses AUD-01. AI audit practice context uses AUD-05 and AUD-06. AI governance context uses AI-01. Attestation boundary context uses BOUND-03. R4B/R1 AARM baseline and GAIC source truth provide the model structure.

Author Synthesis Note

AARM is author synthesis in this paper. It is a readiness vocabulary, not an external maturity model or professional assurance framework.

Candidate Prose

Once evidence objects are defined, organizations need a way to discuss readiness. AARM provides that vocabulary. It asks whether agentic lifecycle work can be reconstructed, tested, evidenced, and reviewed across the object areas this paper has defined.

AARM deliberately avoids scoring. Scores invite overprecision and can be mistaken for certification, compliance grades, vendor rankings, or procurement guidance. AARM instead uses dimensions and levels. The dimensions ask what must be reviewable. The levels describe the degree to which the lifecycle work is observable, trace-linked, evidence-structured, auditability-ready, or assurance-ready for planning.

The word "Assurance-Ready" is sensitive. In AARM, it does not mean assurance has been issued. It means the evidence architecture may be mature enough to support qualified assurance planning or review within a defined scope. It still depends on professional scope, criteria, independence, engagement acceptance, and judgment, and it does not guarantee a positive result.

The ten AARM dimensions are:

1. Audit object clarity.

2. Authority traceability.
3. Human / agent responsibility mapping.
4. Evidence sufficiency.
5. Agent/tool partitioning.
6. Exception traceability.
7. Accepted outcome evidence.
8. Remediation closure.
9. Privacy and selective disclosure.
10. Third-party review boundary.

The levels describe readiness states. L0 Unobservable means lifecycle work cannot be reconstructed beyond informal narrative or final output. L1 Log-Visible means logs exist but responsibility linkage is weak. L2 Trace-Linked means traces connect some actions to agents, tools, or workflows. L3 Evidence-Structured means lifecycle evidence objects exist. L4 Auditability-Ready means defined-scope evidence chains can reconstruct authority, responsibility, tools, outcomes, exceptions, privacy, and closure. L5 Assurance-Ready means evidence architecture may support professional assurance planning or review within a defined scope.

The main use of AARM is conversation discipline. It helps technology leaders avoid saying "we have logs, so we are audit-ready." It helps governance teams ask whether accepted outcomes and remediation closure are evidence states. It helps audit teams separate raw observability from reconstructable evidence. It helps privacy teams see whether selective disclosure is built in. It helps third-party reviewers clarify what their review does and does not mean.

Table 9: Auditability Readiness Levels

Level	Definition	Observable traits	Minimum evidence	What it does not prove	Source / synthesis note
L0 Unobservable	Lifecycle work cannot be reconstructed beyond informal narrative or isolated outputs	Missing lifecycle IDs, no authority linkage, no role map, no accepted	None beyond ad hoc notes or final output	Absence of risk, absence of non-compliance, or system safety	Author synthesis grounded in AUD-01/AU 02

		outcome state			expectation
L1 Log-Visible	Activity logs exist without responsibility-linked evidence chains	Logs, traces, timestamps, tool events	Raw logs or trace exports	Authority, responsibility, acceptance, exception closure, audit readiness	EVID-02/ EVID-03 + author synthesis in this paper
L2 Trace-Linked	Logs/traces link some actions to agents, tools, users, or workflows	Workflow IDs, agent/tool traces, partial actor linkage	Trace-linked run records and partial evidence map	Complete lifecycle responsibility or sufficient evidence	EVID-01/ EVID-02 + author synthesis in this paper
L3 Evidence-Structured	Lifecycle evidence is organized into objects that support reconstruction	Role maps, authority objects, accepted outcome records, partitioning, exception records	MRO-aligned evidence objects and sampled walkthroughs	Assurance readiness, legal compliance, operating effectiveness	GAIC-derivation + AUD-01/AUD-04/AUD-05
L4 Auditability-Ready	Evidence chains can reconstruct defined-scope agentic lifecycle work	Walkthroughs, evidence request response, privacy controls, closure evidence	Complete evidence chain for defined scope plus boundary statement	Audit opinion, certification, regulator approval, legal compliance	author synthesis in this paper grounded in audit/provenance/privacy sources
L5 Assurance-Ready	Evidence architecture may	Repeatable evidence	Reviewable evidence	Actual assurance	author synthesis in

	support	identity,	packs,	opinion,	
	professional	reviewer	integrity	certification,	this paper
	assurance planning	scope,	records,	guaranteed	bounded by
	or review within a	remediation/re	boundary	compliance,	AUD-03/BO
	defined scope	check	statement	regulator	ND-03
		procedure		acceptance	

Table note: AARM is author synthesis grounded in source language. It is not a score, standard, certification, assurance conclusion, procurement benchmark, or vendor ranking.

AARM Dimension Summary

Dimension	Audit question	Minimum evidence family	Primary MRO links	Boundary
Audit object clarity	Can reviewers name the lifecycle unit under review?	Work unit ID, scope, object inventory	MRO-01, MRO-03, MRO-04	Not legal category
Authority traceability	Was the action authorized under scope?	Authority grant, expiry, escalation	MRO-02, MRO-07	Not legal delegation proof
Human / agent responsibility mapping	Who owned intent, execution, review, acceptance, remediation?	Role map and transfer record	MRO-01, MRO-03, MRO-06	No anthropomorphism
Evidence sufficiency	Can reviewers reconstruct work without narrative memory?	Evidence chain and pointers	MRO-04, MRO-08, MRO-16	Not universal sufficiency
Agent/tool partitioning	Can actions be separated across agents/tools/vendors/	Partition keys, tool evidence, substitution	MRO-05, MRO-08, MRO-14,	No vendor ranking

	projects?	record	MRO-15	
Exception traceability	Can deviation be linked to owner and next action?	Exception record and impact scope	MRO-07, MRO-16	Not legal violation
Accepted outcome evidence	Who accepted or disputed the result?	Outcome state, reviewer role, evidence link	MRO-04	Not compliance proof
Remediation closure	What was remediated and who accepted closure?	Corrective action, recheck, closure state	MRO-16	Not settlement
Privacy and selective disclosure	Can review occur without unnecessary exposure?	Redaction, retention, disclosure profile	MRO-10 to MRO-13	Not legal advice
Third-party review boundary	What can a reviewer inspect and claim?	Scope, ruleset, verdict, boundary statement	MRO-11, MRO-12, MRO-16	No certification/assurance opinion

Cross-Links

- Appendix D provides the expanded readiness matrix.
- Chapter 14 connects AARM to enterprise readiness.
- Chapter 15 connects AARM to professional use boundaries.

Boundary Note

AARM is a readiness model only. It does not prove legal compliance, issue assurance, certify systems, rank vendors, bind regulators, or require MPLP.

14. Enterprise Implementation: CIO / CTO / CCO Readiness

Candidate source status: R4 internal candidate source

Chapter purpose: Connect this paper to enterprise readiness without replacing Guide 1 or

Guide 2.

Reader question: What must leaders prepare before auditability or assurance conversations become credible?

Key Claims

- Audit-ready agentic systems require architecture, governance, evidence ownership, privacy controls, exception closure, and role clarity.
- CIO/CTO readiness focuses on technical evidence architecture.
- CCO readiness focuses on policy-to-evidence governance.
- This paper bridges to Guide 1 and Guide 2 but does not replace them.

Source Grounding Note

Enterprise control context uses AUD-04 and AUD-07. AI governance context uses AI-01 and PRIV-03. Big Four sources BF-02 and BF-04 provide market context only. GAIC source truth provides Guide 1/Guide 2 boundaries.

Author Synthesis Note

The CIO/CTO/CCO readiness split is author synthesis in this paper derived from R4C Guide planning and audit evidence architecture in this paper.

Candidate Prose

Enterprises do not become audit-ready by exporting logs at the end of an agentic workflow. Auditability must be designed into architecture, governance, review routines, and evidence ownership. This paper is not an implementation guide, but it defines the readiness questions leaders should ask before Guide 1 and Guide 2 translate the framework into practice.

For CIOs and CTOs, the first question is whether the system can produce evidence by design. Agent runtimes, orchestration layers, tool integrations, workflow engines, ticketing systems, approval systems, data stores, and logging platforms must be able to preserve the relationships that matter: work unit, authority, role, tool action, evidence pointer, outcome, exception, privacy treatment, and closure. If these relationships are not captured at runtime, auditability becomes after-the-fact reconstruction.

The second technology question is partitioning. Evidence should be separable by agent, tool, role, vendor, project, data class, lifecycle state, and review purpose. This requires stable

identifiers, metadata, access controls, retention logic, evidence export, and selective disclosure. It may also require integration with existing observability, security, data governance, and workflow systems.

The third technology question is change. Agentic systems change as models, prompts, tools, vendors, runtimes, policies, workflows, and memory stores change. Auditability requires substitution records, evidence continuity checks, regression evidence, reauthorization triggers, and revalidation boundaries. A system that loses evidence continuity after a runtime change is not auditability-ready for that scope.

For CCOs and governance leaders, the first question is policy-to-evidence translation. Policies should not only describe desired behavior; they should identify the evidence object that proves the policy was operationalized. A delegated authority policy should map to authority records. A human oversight policy should map to role and confirmation records. A remediation policy should map to closure evidence.

The second governance question is role operating model. The IIA Three Lines context is useful here as governance vocabulary, not as a this paper mandate. Management, risk/compliance, internal audit, privacy, security, legal, and technology teams need different evidence views and responsibilities. this paper helps name the evidence layer they must share.

The third governance question is review cadence. Agentic auditability is not a one-time readiness exercise. Evidence objects should be sampled, walked through, rechecked after change, reviewed after exceptions, and re-evaluated when authority, vendor, model, runtime, or privacy conditions change.

Guide 1 will later turn these questions into technical architecture planning. Guide 2 will later turn them into compliance operating-model routines. This paper should not absorb those details. It should define the evidence object layer that both guides will implement.

Enterprise Readiness Crosswalk

Readiness area	CIO / CTO question	CCO / governance question	Evidence object	Guide relation	Boundary
Work unit identity	Can systems assign stable	Which work types require	Work unit record	Guide 1 architecture;	Not audit

	lifecycle IDs?	evidence?		Guide 2 policy scope	procedure
Authority	Can delegated authority be enforced and logged as business scope?	Which policies define authority, expiry, confirmation, and escalation?	Delegated authority record	Guide 1 controls; Guide 2 governance	Not legal delegation proof
Role mapping	Can human/agent roles be captured separately?	Who owns intent, review, acceptance, and closure?	Responsibility map	Both guides	Not legal liability assignment
Tool actions	Can consequential tool actions generate evidence?	Which tool actions require review or confirmation?	Tool-action evidence	Guide 1 high relevance	No liability conclusion
Privacy	Can evidence be minimized, redacted, retained, and selectively disclosed?	Which privacy rules and approvals govern review?	Disclosure and retention profile	Both guides	No legal advice
Exceptions	Can exceptions link to lifecycle objects and closure?	Who governs dispute/remediation on workflow?	Exception/remediation record	Both guides	No legal remedy claim
Third-party review	Can evidence be exported/replayed safely?	What can third parties claim or not claim?	Validation scope and boundary statement	Guide 1 export; Guide 2 review boundary	No certification

Table note: Author-synthesis crosswalk grounded in controls, governance, privacy, and GAIC

Guide planning. It is not procurement guidance or readiness guarantee.

Cross-Links

- Guide 1 will own technical implementation details.
- Guide 2 will own compliance operating-model details.
- Appendix D can be used as a readiness conversation input.

Boundary Note

This chapter does not claim that following this paper makes an enterprise compliant, audit-ready in a professional engagement, certified, regulator-approved, or procurement-preferred.

15. How Audit and Assurance Firms Can Use This Framework

Candidate source status: R4 internal candidate source

Chapter purpose: Define cautious professional use without claiming endorsement, replacement methodology, assurance opinion, or standard-setting status.

Reader question: How might audit and assurance firms use this paper responsibly?

Key Claims

- Audit and assurance firms may use this paper as a conceptual object/evidence request and readiness framing tool.
- This framework can support discussion, scoping, evidence request design, walkthrough planning, and methodology exploration.
- This paper does not replace professional methodology, engagement criteria, independence, judgment, standards, or reporting.

Source Grounding Note

Big Four context uses BF-02 and BF-03 as market context only. Assurance boundary uses AUD-03 and BOUND-03. AI audit practice context uses AUD-05 and AUD-06.

Author Synthesis Note

The professional-use framing is author synthesis in this paper. It is intentionally cautious and non-endorsement based.

Candidate Prose

This framework is relevant to audit and assurance firms because agentic AI changes the evidence question. Firms already discuss AI assurance, trusted AI, audit transformation, technology-enabled assurance, governance, controls, and responsible AI. Agentic systems add another layer: lifecycle work that must be reconstructed across authority, responsibility, agents, tools, outcomes, exceptions, privacy, and remediation.

The safest professional use of this paper is as a framing tool. It can help teams ask what the review object is. It can help distinguish logs from audit evidence chains. It can help convert governance concerns into evidence requests. It can help structure walkthrough conversations. It can help identify whether an enterprise has evidence gaps before any formal assurance claim is discussed.

This paper may also help methodology teams think about agentic evidence architecture. A method team could use the Agentic Audit Object as a conceptual lens: What lifecycle work unit is under review? What authority record exists? What human role accepted the outcome? What tool-action evidence exists? What exception state or remediation closure exists? What privacy treatment governs disclosure?

Internal audit teams may use this paper differently. They may use it to scope readiness reviews, design evidence request lists, test walkthrough templates, ask whether human oversight is evidence-backed, or evaluate whether remediation closure is reconstructable. Again, this use is readiness-oriented. It does not transform this paper into an internal audit standard.

Assurance teams must be especially careful. AARM's "Assurance-Ready" level does not mean assurance has been issued or is guaranteed. It means that an evidence architecture may be mature enough to support qualified assurance planning or review, subject to the firm's professional standards, criteria, engagement scope, independence, risk assessment, evidence evaluation, and reporting requirements.

This paper should never be represented as a Big Four methodology. It should never be described as endorsed by Deloitte, PwC, EY, KPMG, ISACA, IIA, IAASB, PCAOB, AICPA, ISO, NIST, OECD, EU institutions, Singapore agencies, ICO, or any source organization. Sources provide context and terminology. They do not approve this framework.

Professional Use Matrix

Use case	This framework can	this paper cannot	Useful objects	Boundary
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	support	replace		
Readiness conversation	Shared vocabulary for auditability gaps	Professional judgment or engagement acceptance	AARM, evidence chain, object model	No assurance conclusion
Evidence request design	Categories for authority, roles, tools, outcomes, exceptions, privacy, closure	Audit procedures or legal demands	Appendix A request list	Not formal procedure
Walkthrough scoping	Lifecycle reconstruction pattern	Sampling methodology or engagement workplan	Appendix B walkthrough	Illustrative only
Internal audit review	Auditability gap identification	Internal audit standards or final reports	MRO mapping, closure checklist	No standard-setting
Methodology exploration	Object model for agentic work	Firm methodology or regulatory requirements	Agentic Audit Object, Audit Evidence Chain	No Big Four endorsement
Assurance planning	Evidence architecture discussion	Assurance criteria, independence, evidence evaluation, opinion	AARM L4/L5 boundary	No assurance opinion

Table note: Professional-use matrix is author synthesis grounded in assurance and AI audit practice sources. It is not a firm methodology, engagement template, or endorsement claim.

Cross-Links

- Chapter 1 defines scope and boundary.
- Chapter 12 defines third-party validation boundaries.
- Appendix F provides safe language.

Boundary Note

This chapter does not claim that audit or assurance firms need this paper, endorse this paper, have adopted this paper, or should replace their methods with this paper.

16. Conclusion

Candidate source status: R4 internal candidate source

Chapter purpose: Close this paper's argument and bridge to Guide 1, Guide 2, and the later insurability white paper without publication or insurability overclaim.

Reader question: What must exist before agentic assurance and insurability conversations can mature?

Key Claims

- Auditability begins when agentic lifecycle work becomes reconstructable.
- Logs record activity; evidence chains support responsibility review.
- This paper prepares the path for Guide 1, Guide 2, and the later insurability white paper.

Source Grounding Note

Audit evidence and assurance boundary language uses AUD-01 and AUD-03. AI governance context uses AI-01. GAIC source truth supports the Global AI Compliance White Paper 2026 / this paper / later insurability white paper object-chain relationship and companion-paper boundary.

Author Synthesis Note

The Compliance Object -> Audit Evidence Object -> Insurable Risk Object chain is GAIC/author synthesis in this paper. The later insurability white paper source research is deferred.

Candidate Prose

Agentic AI changes the auditability problem because agentic systems do not merely produce outputs. They perform lifecycle work. They act through roles, tools, delegated authority, handoffs, privacy boundaries, exceptions, and remediation workflows. If that work cannot be reconstructed, the system may be impressive, observable, and even well-governed in parts, but it is not audit-ready in the sense this paper defines.

The core thesis therefore remains simple: AI agent auditability cannot be built on logs alone. Logs record activity; evidence chains support responsibility review. The difference is the lifecycle responsibility layer. A log may show that an event occurred. An Audit Evidence Chain shows how that event relates to authority, human role, agent role, tool action, evidence pointer, accepted outcome, exception, privacy treatment, and closure.

This paper's contribution is the Audit Evidence Object layer. The Global AI Compliance White Paper 2026 defined the compliance object layer through Agentic Lifecycle Governance and MROs. This paper translates that layer into evidence requests, walkthroughs, partitioning rules, privacy-aware disclosure, validation boundaries, and readiness levels. It defines Agentic AI Auditability as the ability to reconstruct, test, and evidence agentic lifecycle work across authority, responsibility, tools, outcomes, exceptions, and remediation.

The paper also sets limits. It does not issue assurance. It does not certify systems. It does not prove legal compliance. It does not bind regulators. It does not replace professional audit or assurance methodology. It does not rank vendors. It does not require MPLP. It does not make Validation Lab a certification authority.

These limits are not disclaimers at the edge of the work; they are part of the architecture. Auditability is strongest when the evidence object and the claim boundary are both explicit.

For enterprises, the next step is design. CIOs and CTOs need evidence architecture that captures lifecycle responsibility by design, not after-the-fact logs by accident. CCOs and governance leaders need policy-to-evidence operating models that turn authority, oversight, privacy, exception, and remediation policies into reviewable objects. Internal audit and risk teams need walkthrough structures that test reconstructability before assurance claims appear.

For audit and assurance firms, the next step is method dialogue. This framework can support object-model discussions, readiness scoping, evidence request framing, and lifecycle walkthrough design. It cannot replace professional standards, engagement criteria, independence, evidence evaluation, or reporting.

For the series, this paper completes the second layer of the 3+3 front half. The Global AI Compliance White Paper 2026 established the compliance object layer. This paper establishes the audit evidence object layer. Guide 1 will translate auditability into technical architecture. Guide 2 will translate auditability into compliance operating model. The later insurability white paper will later ask what evidence and governance structures are needed before insurable risk

can be evaluated.

The object chain is:

Series layer	Object layer	Question	Boundary
Global AI Compliance White Paper 2026	Compliance Object	What lifecycle responsibility objects does agentic governance require?	Not legal compliance proof
Agentic AI Auditability & Assurance White Paper 2026	Audit Evidence Object	Can agentic lifecycle work be reconstructed through responsibility-linked evidence chains?	Not audit standard, certification, or assurance opinion
Later insurability white paper	Insurable Risk Object	What risk evidence may later support insurability analysis?	Deferred; no insurance guarantee
Guide 1	Technical implementation path	How can systems capture audit-ready evidence by design?	Future guide, not drafted here
Guide 2	Compliance operating model path	How can governance translate policy into evidence?	Future guide, not drafted here

Table note: GAIC/author-synthesis table. It preserves companion-paper sequencing and does not claim conclusions for the later insurability white paper.

Auditability begins when agentic lifecycle work becomes reconstructable. Logs record activity; evidence chains support responsibility review. This paper prepares the path for Guide 1, Guide 2, and the later insurability white paper by defining the evidence object layer that must exist before agentic assurance or insurability conversations can be serious.

Cross-Links

- Appendix F preserves conclusion-safe boundary language.

- Guide 1 and Guide 2 remain future practitioner guides.
- The later insurability white paper remains a future white paper with separate source research.

Boundary Note

This chapter does not claim this paper is published, final, sealed, regulator-approved, adopted by firms, or sufficient for assurance, certification, legal compliance, procurement, or insurance.

Appendix A — Agentic Audit Evidence Request List

Candidate source status: R4 publication-candidate preparation appendix

Purpose: Provide a structured evidence request catalog for agentic systems.

Intended reader: Audit/assurance teams, internal audit, AI governance, CIO/CTO/CCO teams, technology risk teams, and evidence architecture owners.

Source Grounding

Audit evidence and control vocabulary: AUD-01, AUD-02, AUD-04, AUD-05, AUD-06.

GAIC dependencies: MRO-01 through MRO-16, Evidence-Based Validation Pattern, Validation Lab boundary.

Boundary

This appendix is an evidence request architecture. It is not a formal audit procedure, legal demand, certification criterion, procurement checklist, assurance engagement plan, or proof of legal compliance.

Evidence Request Catalog

Request ID	Lifecycle stage	Evidence requested	Related MRO	Owner	Privacy treatment	Boundary note
ER-01	Work initiation	Work unit ID, task intent, request source, scope, excluded	MRO-08	Request owner	Record only necessary purpose/context fields	Does not prove business appropriateness

		scope				
		Human role for intent, authorization, review, acceptance, exception, remediation				
ER-02	Role mapping		MRO-01	Governance/control owner	Role data minimized to function and authority	Does not assign legal liability
		Authority grant, conditions, expiry, risk class, confirmation and escalation rules				
ER-03	Delegated authority		MRO-02	Delegating role	Restrict authority evidence to review scope	Not legal delegation proof
		Agent role, capability boundary, constraints, evidence duties, escalation route				
ER-04	Agent role		MRO-03	Platform owner	Avoid storing unnecessary prompt/user data	Agent is not a human role
		Acceptance criteria, reviewer role, accepted/rejected/disputed/remediated state				
ER-05	Accepted outcome		MRO-04	Review owner	Redact sensitive output fields where possible	Acceptance is not compliance proof
		Tool ID, action type, target				
ER-06	Tool action		MRO-05	Tool-action	Classify affected data and redact as	Not legal liability

		system, affected data/process, reversibility, rollback path		owner	needed	finding
ER-07	Responsibility transfer	Source agent, target agent, transferred scope, retained scope, inherited constraints	MRO-06	Process owner	Partition by work unit and agent role	Not legal responsibility transfer
ER-08	Authority drift	Baseline authority, observed action, drift trigger, escalation, stop/downgrade/reauthorization	MRO-07	Escalation owner	Limit exposure to event-relevant fields	Not every drift is legal breach
ER-09	Evidence partitioning	Partition keys, links, evidence pointers, integrity hashes, access/disclosure rules	MRO-08	Evidence owner	Use scoped access and redaction profiles	Not blanket retention
ER-10	Cross-project reuse	Source context, target context, reset	MRO-09	Reuse owner	Revalidate data categories and	Not legal reuse clearance

		validation, privacy review, authorization refresh			retention	
ER-11	Privacy lifecycle	Data flow, processing purpose, retention rule, rights workflow, minimization treatment	MRO-10	Privacy owner	Apply privacy-by-design review	Not legal advice
ER-12	Third-party validation	Ruleset identity, evidence pointer, disclosure/redaction profile, verdict hash, reviewer boundary	MRO-11	Validation/ review owner	Use selective disclosure package	Not certification
ER-13	Evidence minimization	Minimization rule, disclosure justification, redaction profile, access log, expiry	MRO-12	Evidence/ privacy owner	Minimize and document residual exposure	Not privacy compliance proof
ER-14	Rights vs retention	Request type, retention basis, hold status, deletion/redaction action,	MRO-13	Privacy/legal owner	Requires jurisdiction-specific review	Not legal interpretation

		review owner			
ER-15	Processor chain	Processor role, subprocessor chain, data processing pointer, responsibility owner	MRO-14	Vendor/ privacy owner	Restrict processor evidence by need-to-know Not vendor ranking
ER-16	Substitution conformance	Prior/new model, vendor, runtime, tool, evidence-integrity check, reauthorization	MRO-15	Change owner	Preserve evidence continuity without over-retention Not conformance certification
ER-17	Incident/ remediation closure	Event, affected outcome, owner, corrective action, recheck, closure acceptance, reopen criteria	MRO-16	Remediation owner	Redact sensitive incident details as needed Not legal closure

Guide Relationship

Guide 1 should translate this appendix into system output requirements, evidence export design, runtime metadata, and workflow architecture. Guide 2 should translate it into policy-to-evidence governance, audit preparation, review cadence, and closure routines.

Placement Recommendation

Retain a condensed version in the main artifact. A fuller operational checklist can be deferred to Guide 1 and Guide 2.

Appendix B — Agentic System Walkthrough Template

Candidate source status: R4 publication-candidate preparation appendix

Purpose: Provide a repeatable walkthrough structure for a single agentic lifecycle run.

Intended reader: Internal audit, technology risk, AI governance committees, architecture review teams, platform owners, and compliance engineering teams.

Source Grounding

Control and audit practice context: AUD-04, AUD-05, AUD-06.

Provenance and observability context: EVID-01, EVID-02.

GAIC dependencies: MRO-01 through MRO-08 and MRO-16.

Boundary

This template is illustrative. It is not a formal audit procedure, assurance engagement program, legal compliance test, certification path, or regulator-approved walkthrough.

Walkthrough Fields

Field	Description	Evidence pointer	Related MRO	Boundary note
Work unit ID	Stable identifier for the lifecycle work under review	Workflow/run/task record	MRO-08	Not mandatory schema
Intent and scope	Purpose, initiating request, in-scope and out-of-scope actions	Request record, policy reference	MRO-01, MRO-08	Does not prove appropriateness
Initiator role	Human or system role that initiated the work	Role map	MRO-01	Not legal liability assignment
Delegated authority	Authority grant, conditions, expiry,	Authority record	MRO-02	Not legal delegation proof

confirmation, escalation

Agent role	Agent identity, role boundary, constraints, capability limits	Agent registry/config record	MRO-03	Agent role is not human role
Tool/system used	Tool, external system, API, processor, runtime	Tool-action evidence	MRO-05, MRO-14	Not vendor ranking
Evidence pointers	Logs, traces, approvals, tickets, records, hash manifests	Evidence chain references	MRO-08, MRO-12	Not blanket disclosure
Human review	Reviewer role, review criteria, review result	Review record	MRO-01, MRO-04	Not assurance opinion
Accepted outcome	Produced/accepted/rejected/disputed/remediated state	Outcome record	MRO-04	Not compliance proof
Exception state	Drift, dispute, failure, privacy event, tool side effect	Exception record	MRO-07, MRO-16	Not legal breach finding
Remediation closure	Corrective action, recheck, closure acceptance, reopen criteria	Remediation record	MRO-16	Not legal settlement
Privacy treatment	Data category, redaction, retention, disclosure profile	Privacy/disclosure record	MRO-10 to MRO-13	Not legal advice
Third-party review boundary	Reviewer role, ruleset, scope, result, limitation	Validation/review record	MRO-11	Not certification

Walkthrough Sequence

1. Identify the work unit and review scope.
2. Confirm the authority boundary and role ownership.
3. Reconstruct agent actions, tool actions, and handoffs.
4. Link logs/traces to evidence pointers and lifecycle objects.
5. Verify accepted outcome, dispute, or rejection state.
6. Review exceptions and remediation closure.
7. Confirm privacy/selective disclosure treatment.
8. Record what the walkthrough supports and what it does not prove.

Guide Relationship

Guide 1 should use this template as a technical evidence export and walkthrough architecture pattern. Guide 2 should use it as an audit preparation and governance review workflow.

Placement Recommendation

Include in the main artifact if space allows. Expand implementation details in Guide 1 and governance workflow details in Guide 2.

Appendix C — MRO-to-Audit-Evidence Mapping

Candidate source status: R4 publication-candidate preparation appendix

Purpose: Preserve the full mapping from GAIC MROs to audit evidence objects in this paper.

Intended reader: Audit/assurance method teams, compliance engineering, internal audit, AI governance, Guide 1/2 authors.

Source Grounding

GAIC source truth defines MRO-01 through MRO-16. Audit evidence and control language is grounded in AUD-01, AUD-05, and AI-01. This appendix is GAIC-derived plus author synthesis in this paper.

Boundary

MROs are proposed GAIC governance objects. They are not legal mandates, audit standards, certification criteria, procurement criteria, or legal compliance proof.

Full Mapping

MRO	Evidence object	Audit question	Evidence request	AARM dimension	Guide 1 relation	Guide 2 relation
MRO-01 Human Role to MAS Responsibility Mapping	Human-role responsibility map	Who owned intent, authorization, review, acceptance, exception, and remediation?	Role map with lifecycle stage and evidence duty	Human / agent responsibility mapping	Capture roles in workflows and evidence metadata	Define responsibilities, accountabilities, and routines
MRO-02 Delegated Authority Boundary	Delegated authority record	Was the agent authorized for the action under scope and condition?	Authority grant, expiry, risk class, escalation, reauthorization	Authority traceability	Implement authority checks and evidence records	Governance, delegations, policy and approvals
MRO-03 Agent Role is not Human Role	Agent role boundary object	Was the agent role separated from human accountability?	Agent role, constraints, owner, escalation	Human / agent responsibility mapping	Define agent role registry and capability boundaries	Governance, role owners
MRO-04 Accepted Outcome Compliance	Accepted outcome record	Was the output accepted, rejected, disputed, or remediated?	Acceptance criteria, reviewer, outcome state, evidence link	Accepted outcome evidence	Capture outcome state transitions	Governance, review and acceptance
MRO-05 Tool-Action Liability	Tool-action evidence object	What tool created	Tool, target, action,	Agent/tool partitioning	Instrument tool-action	Define tool action

Boundary		consequence, under what authority, and with what reversibility?	affected data, rollback, owner		evidence	approval review
MRO-06 Responsibility Transfer Across Agents	Responsibility transfer record	Did responsibility transfer or remain with source role?	Source/ target agent, transferred/retained scope, inherited constraints	Human / agent responsibility mapping	Capture handoff records	Governance transfer escalation
MRO-07 Authority Drift	Authority drift exception record	Did observed action exceed delegated authority?	Baseline, observed action, trigger, escalation, closure	Authority traceability; exception traceability	Detect and record drift	Governance escalation
MRO-08 MAS Evidence Partitioning	Partitioned evidence chain	Can evidence be separated and linked for review?	Partition keys, links, hashes, evidence pointers, access rules	Evidence sufficiency; agent/tool partitioning	Build partitioned evidence architecture	Governance evidence access
MRO-09 Cross-Project Reuse Compliance	Reuse context validation record	Was reused agent/workflow context reset and revalidated?	Source/ target context, reset validation, privacy review, authorization	Agent/tool partitioning	Implement reuse validation checks	Governance policy

			refresh			
MRO-10 Privacy / GDPR Lifecycle Mapping	Privacy lifecycle evidence object	How does lifecycle work map to data processing and retention?	Data flow, purpose, retention rule, rights workflow, minimization	Privacy and selective disclosure	Capture privacy metadata	Govern privacy review
MRO-11 Privacy- Preserving Third- Party Validation	Validation disclosure profile and verdict record	Can third-party review happen without raw over- disclosure?	Ruleset, redaction/dis closure profile, evidence pointer, verdict hash	Third-party review boundary; privacy	Support export/repla y packages	Govern party re
MRO-12 Evidence Minimization and Selective Disclosure	Evidence minimization and disclosure object	Is disclosed evidence limited to review purpose?	Minimization rule, disclosure justification, redaction profile, access log	Privacy and selective disclosure	Build redaction/s elective disclosure	Govern disclosu approva
MRO-13 Data Subject Rights vs Evidence Retention	Rights-retention reconciliation record	How are rights requests reconciled with evidence retention?	Request, retention basis, hold status, redaction/del etion action	Privacy and selective disclosure	Support retention/rig hts workflows	Govern legal/pr review
MRO-14 Third- Party Processor / Subprocessor Chain	Processor/ subprocessor chain evidence	Which third parties processed or affected	Processor role, subprocesso r chain, data	Agent/tool partitioning	Capture vendor/proc essor	Govern process chain

			lifecycle work?	agreement pointer, responsibilit y owner	evidence	
MRO-15 Vendor / Model / Runtime Substitution Conformance	Substitution conformance record	Did evidence continuity survive component change?	Prior/new component, integrity check, reauthorizati on, revalidation	Agent/tool partitioning; evidence sufficiency	Implement substitution evidence controls	Govern change review
MRO-16 Incident, Dispute, and Remediation Closure	Incident/ remediation closure record	Did exception or dispute close with evidence?	Event, affected outcome, owner, corrective action, recheck, closure	Exception traceability; remediation closure	Build closure workflow evidence	Govern remedia closure

Guide Relationship

Guide 1 should treat this appendix as an implementation-neutral architecture checklist. Guide 2 should treat it as a policy-to-evidence and compliance operating model checklist.

Placement Recommendation

Include full mapping in the this paper appendix. Consider a shorter summary table in Chapter 6.

Appendix D — Auditability Readiness Matrix

Candidate source status: R4 publication-candidate preparation appendix

Purpose: Provide AARM dimension and readiness-level crosswalk.

Intended reader: Audit readiness teams, CIO/CTO/CCO leaders, internal audit, governance

committees, platform teams.

Source Grounding

Audit evidence: AUD-01.
AI audit practice: AUD-05, AUD-06.
AI governance: AI-01.
Boundary: BOUND-03.
GAIC dependency: R4B/R1 AARM baseline, MROs, ALCS context, Validation Lab boundary.

Boundary

AARM is not an audit standard, certification, assurance opinion, legal compliance proof, regulator approval, procurement recommendation, vendor ranking, or score.

AARM Dimensions

Dimension	Definition	Audit question	Evidence expected	GAIC/MRO dependency	Author-synthesis note	Boundaries
Audit object clarity	System can identify lifecycle work under review	Can reviewers name the lifecycle unit, scope, object and outcome?	Lifecycle ID, workflow scope, object inventory	MRO-01, MRO-03, MRO-04	AARM synthesis	Not
Authority traceability	Delegated authority can be traced to action	Was the action authorized within scope?	Authority grant, expiry, escalation, reauthorization	MRO-02, MRO-07	AARM synthesis	Not dele
Human / agent responsibility mapping	Human and agent roles are separated and mapped	Who owned intent, execution,	Role-responsibility map, transfer record	MRO-01, MRO-03, MRO-06	AARM synthesis	No antl

		review, acceptance , remediation?				
Evidence sufficiency	Evidence supports reconstruction under review scope	Can reviewers reconstruct without narrative memory?	Evidence chain, pointers, integrity records	MRO-04, MRO-08, MRO-16	AARM synthesis	Not suffi con
Agent/tool partitioning	Evidence separates agent, tool, vendor, processor, project	Can reviewers identify which surface produced each evidence item?	Agent/tool trace, partition keys, substitution record	MRO-05, MRO-08, MRO-14, MRO-15	AARM synthesis	No
Exception traceability	Exceptions link to lifecycle objects and owners	Can deviation be linked to authority, impact, and next action?	Exception record, impact scope, escalation	MRO-07, MRO-16	GAIC-derived synthesis	Not find
Accepted outcome evidence	Outcome states are recorded with role and evidence linkage	Who accepted, rejected, disputed, or	Outcome state, reviewer role, evidence link	MRO-04	GAIC-derived synthesis	Acco con

		remediated				
		the result?				
		What was				
Remediation	Corrective action	remediated	Remediation plan,			
closure	and closure are	and who	correction	MRO-16	GAIC-derived	Not
	evidence-backed	accepted	evidence, recheck,		synthesis	sett
		closure?	closure status			
		Can review				
Privacy and	Evidence can be	happen	Redaction,			
selective	minimized and	without	retention,	MRO-10 to	author synthesis in	Not
disclosure	disclosed by	exposing	disclosure profile	MRO-13	this paper	Not
	scope	more than				
		needed?				
Third-party	Review role and	What can	Scope, ruleset,	MRO-11,		
review	claim boundary	reviewers	verdict, boundary	MRO-12,	author synthesis in	No
boundary	are explicit	inspect and	statement	MRO-16	this paper	cer
		not claim?				

Readiness Levels

Level	Definition	Observable traits	Minimum evidence	What it does not prove	Source / synthesis classification
L0 Unobservable	Lifecycle work cannot be reconstructed	Outputs only, informal memory, no lifecycle ID	None or ad hoc notes	Absence of risk or non-compliance	author synthesis in this paper grounded in audit evidence concepts
L1 Log-Visible	Activity logs exist but are not responsibility-linked	Logs, traces, timestamps, event streams	Raw logs or trace exports	Authority, responsibility, accepted	EVID-02/ EVID-03 + author

				outcome, closure	synthesis in this paper
L2 Trace-Linked	Logs/traces connect some actions to workflows or actors	Workflow IDs, agent/tool traces, partial actor link	Trace-linked run records	Complete lifecycle responsibility or sufficient evidence	EVID-01/ EVID-02 + author synthesis in this paper
L3 Evidence-Structured	Lifecycle objects exist for reconstruction	Role maps, authority objects, outcome records, exception records	MRO-aligned evidence objects	Assurance readiness, legal compliance, operating effectiveness	GAIC-derive + audit/contr grounding
L4 Auditability-Ready	Defined-scope evidence chains reconstruct lifecycle work	Walkthroughs, evidence requests, privacy controls, closure records	Complete evidence chain for review scope	Audit opinion, certification, regulator approval	author synthesis in this paper
L5 Assurance-Ready	Evidence architecture may support professional assurance planning or review within a defined scope	Repeatable evidence packs, reviewer scope, rulesets, recheck procedures	Reviewable evidence packs and boundary statement	Actual assurance, certification, legal compliance, regulator acceptance	author synthesis in this paper bounded by AUD-03/BO D-03

Minimum Evidence Crosswalk

Dimension	L0/L1 concern	L2/L3 transition	L4/L5 transition
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Audit object clarity	No stable work unit	Work unit IDs appear in traces and evidence objects	Work units support scoped walkthroughs and evidence packs
Authority traceability	Access logs only	Partial authority metadata	Delegated authority records are reconstructable
Responsibility mapping	User/service IDs only	Partial role map	Human/agent responsibility chain is explicit
Evidence sufficiency	Narrative memory	Evidence pointers emerge	Evidence chain supports review objective
Partitioning	Flattened logs	Partial partitions	Agent/tool/vendor/project/privacy partitions are linked
Exception traceability	Incident tickets only	Exceptions linked to work units	Exceptions include owner, action, recheck, closure
Accepted outcome	Output completion only	Partial review records	Accepted/rejected/disputed/remediated states are evidence-backed
Remediation closure	Closed ticket only	Corrective action tracked	Closure is accepted and reopen criteria exist
Privacy/selective disclosure	Collect everything or disclose ad hoc	Redaction/retention rules appear	Selective disclosure packages support review
Third-party boundary	No scope statement	Limited reviewer scope	Ruleset, scope, verdict, and non-certifying boundary are explicit

Appendix E — Exception / Remediation Closure Checklist

Candidate source status: R4 publication-candidate preparation appendix

Purpose: Define closure evidence requirements for exceptions, disputes, and remediation.

Intended reader: Internal audit, incident governance, compliance, risk, security, AI platform owners, remediation owners.

Source Grounding

Incident/governance context: EVID-04, AI-02.

Controls and audit practice: AUD-04, AUD-05, AUD-06.

GAIC dependencies: MRO-04, MRO-07, MRO-16.

Boundary

This checklist frames closure as evidence-backed governance state. It does not prove legal remedy, settlement, regulator acceptance, operational effectiveness, certification, or assurance.

Closure Checklist

Event type	Evidence required	Owner	Corrective action	Closure status	Reviewer	Reopen criteria	Bou not
Authority exception	Baseline authority, observed action, trigger, work unit, evidence pointer	Escalation owner	Stop, downgrade, reauthorize, confirm, or constrain	Open / remediated / closed	Authority reviewer	Repeat drift, scope expansion, failed confirmation	Not breac find

Responsibility gap	Missing role owner, affected lifecycle stage, agent role, work unit	Governance owner	Assign owner, update role map, recheck workflow	Open / assigned / closed	Governance reviewer	Owner missing in later run	Not liable, assets
Tool-action side effect	Tool call, affected system/data, reversibility, rollback evidence	Tool-action owner	Rollback, correction, disable tool, add approval	Open / corrected / closed	System owner	New side effect or rollback failure	Not concerned
Outcome dispute	Output, criteria, dispute reason, reviewer notes	Review owner	Re-review, correct, reject, accept with limitation	Open / disputed / remediated / accepted	Outcome reviewer	New evidence or recurring dispute	Not settled, proceed
Privacy event	Data category, disclosure path, redaction/retention record	Privacy owner	Redact, restrict, delete, hold, notify internal process	Open / contained / closed	Privacy reviewer	Unauthorized exposure or rights request	Not advised
Handoff failure	Source/target agent, transferred scope, retained scope,	Process owner	Return, reassign, clarify constraints	Open / reassigned / closed	Process reviewer	Repeated failed handoff	Not transfer

	rejection reason						
	Source context, target context, reuse decision, missing reset	Reuse owner	Revalidate, reset, withdraw, update evidence	Open / revalidated / closed	Reuse/ governance reviewer	Reuse in new context	Not reus clea
Cross-project reuse issue							
	Prior/new component, evidence break, regression or revalidation status	Change owner	Reauthorize, retest, revalidate evidence chain	Open / revalidated / closed	Change reviewer	Component change or failed evidence continuity	Not com cert
	Prior remediation, failed recheck, affected outcome	Remediation owner	Reopen, assign corrective action, recheck	Reopened / corrected / closed	Closure reviewer	Failed recheck or recurring exception	Not clos

Closure Evidence Fields

- Event ID.
- Work unit ID.
- Affected MRO.
- Affected outcome.
- Owner.
- Evidence pointer.
- Corrective action.

- Recheck evidence.
- Closure reviewer.
- Closure status.
- Reopen criteria.
- Privacy treatment.
- Boundary statement.

Guide Relationship

Guide 1 should translate closure fields into workflow, ticketing, runtime, and evidence-export architecture. Guide 2 should translate closure fields into incident/dispute/remediation governance routines.

Appendix F — Boundary Language

Candidate source status: R4 publication-candidate preparation appendix

Purpose: Provide safe language for auditability, assurance, validation, MRO, AARM, MPLP, privacy, and professional-use boundaries.

Intended reader: Author, editors, reviewers, publication QA, Guide 1/2 teams, source QA.

Source Grounding

Assurance and attestation boundary: AUD-03, BOUND-03.

Conformity/certification boundary: BOUND-01, BOUND-02.

GAIC dependencies: Validation Lab boundary, MPLP boundary, companion-paper boundary, R4A/R4B/R4C boundary QA.

Allowed / Forbidden / Replacement Language

Claim area	Allowed wording	Forbidden wording	Safe replacement
Paper status	"publication-candidate preparation source"; "internal candidate package"; "source-grounded candidate"	"published"; "final"; "sealed"; "live public release"	"R4 publication-candidate preparation package, not public/final/sealed/live"

source"

Auditability	"proposed auditability framework"; "lifecycle evidence guide"	"audit standard"; "official audit methodology"	"auditability readiness framework"
Evidence	"evidence chain supports reconstruction"	"evidence chain proves compliance"	"evidence chain supports responsibility review"
Agentic Audit Object	"proposed object model in this paper"	"required schema"; "legal category"	"conceptual review object"
MRO	"GAIC-derived governance object"	"legal requirement"; "regulator-mandated object"	"proposed lifecycle responsibility object"
AARM	"proposed readiness model"	"certification level"; "maturity score"; "assurance grade"	"readiness vocabulary"
Assurance	"may support assurance planning subject to professional scope"	"assures"; "guarantees assurance"; "issues assurance opinion"	"may support readiness discussion"
Certification	"non-certifying review"; "not certification"	"certified"; "certifies systems"; "certification-ready"	"reviewable under defined scope"
Legal/privacy	"privacy/legal context"; "requires legal review"	"GDPR compliant"; "legal compliance proof"; "legal advice"	"privacy-aware evidence architecture"
Big Four	"reviewed sources create market context"	"Big Four endorse this paper"; "Big Four need this paper"	"Big Four sources discuss related AI assurance/governance topics"

Audit/ professional bodies	"supports terminology or boundary"	"endorses this paper"; "adopts AARM"	"provides external professional-language context"
Validation Lab	"non-certifying evidence adjudication example"	"certification authority"; "proves compliance"; "regulator-approved"	"one optional example of evidence review boundary"
MPLP	"one optional protocol path"	"required"; "industry standard"; "only correct path"	"one possible implementation path"
Cognitive OS	"possible runtime path for evidence capture in related product-stack planning"	"required for auditability"	"one possible implementation environment, outside this paper requirement"
SoloCrew	"product proof path for workflow patterns in planning notes"	"proves enterprise readiness"	"illustrative product proof path, not an enterprise- readiness claim"
SEO/GEO	"future planning note"	"uplift achieved"; "answer-engine recognized"	"future publication/visibility work after artifact exists"
Procurement	"readiness conversation input"	"recommended vendor"; "procurement ranking"	"non-procurement framework"
Later insurability white paper	"future insurability paper"	"insurance guarantee"; "underwriting standard"	"future source research required"

Big Four Boundary

Use:

- "The reviewed Big Four sources discuss AI assurance, trusted AI, governance, risk, controls, audit transformation, and agentic AI adoption."

- "These sources create market context for this paper."

Do not use:

- "Big Four firms endorse this paper."
- "The Big Four need this paper."
- "this paper replaces Big Four methodology."

Audit / Professional Body Boundary

Use professional sources for terminology and boundary. Do not imply that PCAOB, IAASB, AICPA, ISACA, IIA, ISO, NIST, OECD, EU institutions, Singapore agencies, ICO, or any other organization endorses or adopts this paper.

Legal / Privacy Boundary

Use privacy and legal sources as context. State that jurisdiction-specific interpretation belongs to qualified legal/privacy professionals. Do not say any evidence object in this paper proves GDPR, EU AI Act, UK GDPR, or other legal compliance.

Validation Lab Boundary

Validation Lab may be described only as one non-certifying evidence adjudication example. It does not certify systems, issue compliance certificates, provide legal opinions, issue assurance opinions, act as regulator, act as conformity assessment body, prove enterprise readiness, or guarantee insurability.

MPLP Boundary

MPLP may be discussed only as one optional protocol path. This paper does not require MPLP, does not claim MPLP is industry-standard, and does not claim MPLP proves compliance, auditability, assurance, or enterprise readiness.

AARM Boundary

AARM is a proposed readiness model. It does not score vendors, certify systems, issue assurance, prove legal compliance, bind regulators, or provide procurement guidance. L5 "Assurance-Ready" means evidence architecture may support qualified assurance planning or review within a defined scope, subject to professional scope and judgment. It does not mean assurance has been issued.

Source Note and Citation Register

R4 uses source-ID citation notes rather than public footnotes. Source IDs appear in chapter source notes and table source-support columns. The candidate package keeps the full source register and chapter-level citation map as internal package records.

R4 Citation Style Decision

- Internal style for R4: source IDs in text and tables, plus a package-level source register.
- Public style for a later wave: decide whether to convert source IDs into endnotes, footnotes, or an appendix source register.
- Quote guidance: paraphrase by default; avoid direct quotation unless a later citation QA wave approves a short, necessary excerpt.
- Boundary guidance: source presence does not imply endorsement, standard-setting, certification, assurance, legal advice, procurement approval, or regulator approval.

Source Role Summary

- BF sources: market context only.
- AUD sources: audit/assurance terminology and boundary context.
- AI sources: governance context, not legal compliance proof.
- EVID sources: observability, provenance, logs, and evidence-chain distinction.
- PRIV sources: minimization, retention, selective disclosure, and privacy/evidence tension.
- BOUND sources: validation, verification, attestation, certification, and conformity-assessment boundaries.
- GAIC-SOURCE: MROs, RCCS-M/ALCS context, Validation Lab boundary, and companion-paper sequencing.

Artifact Integrity Note

R4 generated an internal candidate source package and checksum file. Candidate HTML/PDF/DOCX artifacts were intentionally deferred because the existing renderer scripts are GAIC public-white-paper specific and direct reuse would risk touching public/sealed GAIC artifact assumptions. Artifact generation is documented in the internal generation plan and the R4 artifact-generation decision report.

Package Source Register

AIAAWP R4 Source Register

Status: R4 publication-candidate preparation source register

Boundary: Sources listed here are used for context, terminology, grounding, or boundary discipline. Inclusion does not imply endorsement of this paper, GAIC, MPLP, AARM, Validation Lab, or Jearon Wong.

Source Use Rules

- Big Four sources are market context only.
- Audit/professional sources support audit evidence, assurance, controls, internal audit, and boundary language.
- AI governance/public guidance sources support governance context, risk management, accountability, documentation, monitoring, and human oversight.
- Evidence/provenance/observability/logging sources support the distinction between logs/traces and responsibility-linked audit evidence chains.
- Privacy sources support minimization, retention, selective disclosure, and evidence/privacy tension.
- Boundary sources support distinctions among validation, assurance, attestation, certification, conformity assessment, and audit opinion.
- GAIC source truth supports MROs, RCCS-M/ALCS context, Evidence-Based Validation Pattern, Validation Lab boundary, and companion-paper boundary.

Register

Source ID	Title	Organization	Source class	Use role	Chapters used
BF-01	AI agents: Scaling faster	Deloitte	Big Four	Agentic AI adoption and enterprise readiness	0, 2, 14,

	than expected		market context	context	15
BF-02	Assurance for AI PwC		Big Four market context	AI assurance market context	0, 1, 14, 15
BF-03	EY Assurance releases new technology capabilities strengthening confidence and trust	EY	Big Four market context	Audit transformation and assurance technology context	1, 15
BF-04	KPMG Trusted AI	KPMG	Big Four market context	Trusted AI, governance, privacy, and accountability context	2, 10, 14
BF-05	AI Agents Move Beyond Experimentation as Leaders Prepare for Competitive Transformation Within 24 Months	KPMG	Big Four market context	Agentic AI adoption and risk context	0, 2, 14, 15
AUD-01	AS 1105: Audit Evidence	PCAOB	Audit/ professional body	Audit evidence terminology and sufficiency/appropriateness framing	0, 3, 4, 5, 6, 7, 13, 16
AUD-02	ISA 500	IAASB	Audit/	Audit evidence and	3, 4, 7, 13

	(Revised), Audit Evidence		professional body	technology-enabled evidence context	
AUD-03	ISAE 3000 (Revised)	IAASB	Audit/ professional body	Assurance engagement boundary	1, 12, 13, 15, 16
AUD-04	Trust Services Criteria	AICPA	Audit/ professional body	Control vocabulary, monitoring, privacy/security, control activities	5, 7, 8, 10, 11, 14
AUD-05	Audit and Assurance Guidance for the NIST Cybersecurity Framework 2.0 and Artificial Intelligence	ISACA	Audit/ professional body	AI audit and assurance practice context	6, 7, 8, 11, 13, 15
AUD-06	Global Technology Audit Guide: Auditing Artificial Intelligence	IIA	Audit/ professional body	Internal audit, AI governance, controls, human oversight	2, 7, 8, 11, 13, 14, 15
AUD-07	The IIA's Three Lines Model	IIA	Audit/ professional body	Governance role separation	1, 14, 15
AI-01	Artificial Intelligence Risk Management Framework 1.0	NIST	Regulator/ public guidance	AI risk management, governance, monitoring,	0, 2, 3, 5, 6, 13, 14, 16

				accountability	
AI-02	NIST AI 600-1 Generative AI Profile	NIST	Regulator/ public guidance	Generative AI risk, incidents, monitoring, documentation	2, 7, 11, 14
AI-03	OECD AI Principles	OECD	Public guidance	Responsible AI accountability and transparency context	1, 2, 14
AI-04	ISO/IEC 42001:2023 AI management system	ISO	Standards body	AI management-system context	2, 13, 14
AI-05	ISO/IEC 23894:2023 AI risk management	ISO	Standards body	AI risk-management context	2, 13, 14
AI-06	Regulation (EU) 2024/1689 Artificial Intelligence Act	European Union	Regulator/ public guidance	AI governance, logging, human oversight, documentation context	1, 2, 4, 7, 14
AI-07	Model AI Governance Framework for Generative AI	AI Verify Foundation / IMDA	Public guidance	Generative AI governance and testing context	1, 2, 7, 14
AI-08	Model AI Governance Framework for Agentic AI	Singapore MDDI / AI Verify Foundation	Public guidance	Agentic AI governance, autonomy, accountability, safeguards	0, 2, 3, 5, 6, 14
AI-09	Guidance on AI and Data	UK ICO	Regulator/ public	AI/privacy context	9, 10, 14

	Protection		guidance		
				Provenance vocabulary	
EVID-01	PROV Overview W3C		Technical documentation	for entities, activities, agents, derivation, responsibility	3, 4, 5, 8, 9, 13
EVID-02	OpenTelemetry Observability Primer	OpenTelemetry	Technical documentation	Logs, metrics, traces, observability context	0, 4, 8, 9, 14
EVID-03	SP 800-92 Guide to Computer Security Log Management	NIST	Public guidance	Log management context	4, 9, 10
EVID-04	Cybersecurity Framework 2.0	NIST	Public guidance	Governance, detection, response, recovery, incident context	4, 11, 14
PRIV-01	GDPR Article 5	European Union	Regulator/ public guidance	Minimization, storage limitation, accountability context	9, 10
PRIV-02	EDPB Guidelines 4/2019 on Article 25	EDPB	Regulator/ public guidance	Data protection by design/default and minimization context	9, 10
PRIV-03	NIST Privacy Framework 1.0	NIST	Public guidance	Privacy risk management and governance context	10, 14
PRIV-04	Data	UK ICO	Regulator/ public	Data minimization	10

	minimisation		guidance	context	
PRIV-05	Storage limitation	UK ICO	Regulator/ public guidance	Retention and storage limitation context	10
BOUND-01	CASCO - Committee on conformity assessment	ISO	Standards body	Conformity-assessment boundary	12, 13
BOUND-02	ISO/IEC 17029:2019 summary	ISO	Standards body	Validation/verification body boundary	12, 13
BOUND-03	Attestation standards	AICPA	Audit/ professional body	Attestation and practitioner-report boundary	12, 13, 15
GAIC-SOURCE	GAIC v0.3.2- FRC-R3 source truth	Jearon Wong / GAIC source master	GAIC-derived source truth	MROs, RCCS-M/ALCS context, Evidence- Based Validation Pattern, Validation Lab boundary, companion- paper boundary	1, 3, 5, 6, 9, 10, 11, 12, 13, 16; appendic es

R4 Source Register Result

PASS - all sources used in the R4 candidate source are drawn from the R0 inventory or GAIC source truth.

R4 Citation Style

R4 uses source IDs in text, tables, and source notes with this register as the package-level lookup. Public endnote or footnote formatting remains a later publication-candidate QA decision.

Package Citation Map

AIAAWP R4 Citation Map

Status: R4 publication-candidate preparation citation map

Boundary: This map supports R4 candidate source review. It does not create final citation formatting, quote source text, or claim endorsement by any source organization.

Chapter Citation Map

Chapter	Source IDs used	Source role	Use placement	Citation risk	R4 review note
0 Executive Summary	BF-01, BF-02, BF-05, AUD-01, AI-01, EVID-02, GAIC-SOURCE	Market context; audit evidence; governance; observability; GAIC source truth	Main argument and source note	Big Four endorsement overclaim	Keep Big Four as context only.
	AUD-03, BOUND-01, BOUND-02, BOUND-03, AUD-07, BF-02, BF-03, GAIC-SOURCE	Boundary, audience, assurance/attestation context	Boundary section and table notes	Treating boundary sources as endorsement	Repeat non-standard and non-certifying boundary.
2 Difference	BF-01, BF-04,	Market and AI	Main text	Dismissing	Maintain

from Model Governance	BF-05, AI-01, AI-02, AI-04, AI-05, AI-08, AUD-06	governance context	and Table 1	model governance	"necessary but insufficient" framing.
3 Audit Object Shift	AUD-01, AUD-02, AI-01, AI-08, EVID-01, GAIC-SOURCE	Audit evidence, governance, provenance, GAIC MROs	Main text	Treating Agentic Audit Object as externally sourced	Label as author synthesis in this paper.
4 Logs vs Evidence Chains	AUD-01, AUD-02, EVID-01, EVID-02, EVID-03, EVID-04, AI-06	Audit evidence, observability, logging, incident/governance context	Main text and Table 2	Saying logs are useless	State logs are useful evidence ingredients.
5 Agentic Audit Object Model	AUD-01, AUD-04, AI-01, EVID-01, GAIC-SOURCE	Evidence, controls, governance, provenance, GAIC objects	Main text and object table	Mandatory schema overclaim	Keep model as proposed architecture.
6 MRO Mapping	GAIC-SOURCE, AUD-01, AUD-05, AI-01	GAIC MRO source truth; evidence/control language	Main text and Table 3	MRO legal mandate overclaim	Keep MROs as GAIC-derived governance objects.
7 Evidence Request List	AUD-01, AUD-02, AUD-04, AUD-05, AUD-06, AI-06	Audit evidence, controls, AI audit practice	Main text and Table 4	Evidence request as formal audit procedure	Mark as readiness/evidence architecture.

8 Lifecycle Walkthrough	AUD-04, AUD-05, AUD-06, EVID-01, EVID-02, GAIC-SOURCE	Controls, audit practice, provenance, traces	Main text and Table 5	Formal audit procedure implication	Keep walkthrough illustrative.
9 Evidence Partitioning	EVID-01, EVID-02, EVID-03, PRIV-01, PRIV-02, PRIV-03, AI-09, GAIC-SOURCE	Provenance/ logging/ privacy/context partitioning	Main text and Table 6	Universal disclosure or retention	Stress scope, access, and minimization.
10 Privacy and Minimization	PRIV-01, PRIV-02, PRIV-03, PRIV-04, PRIV-05, AI-09, GAIC-SOURCE	Privacy and retention context	Main text and Table 7	Legal advice or GDPR proof	Require legal/privacy review in R3.
11 Exception and Closure	EVID-04, AI-02, AUD-04, AUD-05, AUD-06, GAIC-SOURCE	Incident, control, remediation, audit practice	Main text and Table 10	Legal remedy/settle ment overclaim only.	Closure is evidence state
12 Third-Party Validation Boundary	AUD-03, BOUND-01, BOUND-02, BOUND-03, GAIC-SOURCE	Assurance, attestation, conformity, validation boundary	Main text and Table 8	Certification or assurance opinion overclaim	Review every Validation Lab paragraph.
13 AARM	AUD-01, AUD-05, AUD-06, AI-01, BOUND-03, GAIC-SOURCE	Evidence, AI audit practice, governance, attestation boundary, AARM source truth	Main text and Table 9	Maturity/ certification overclaim	Keep "what it does not prove" column.
14 Enterprise	BF-02, BF-04,	Enterprise	Main text	Procurement/	Bridge only to

Readiness	AUD-04, AUD-07, AI-01, PRIV-03, GAIC-SOURCE	governance, controls, three lines, privacy, market context		readiness guarantee	Guide 1/Guide 2.
15 Audit and Assurance Firm Use	BF-02, BF-03, AUD-03, AUD-05, AUD-06, BOUND-03	Market context, professional practice, assurance/atte station boundary	Main text	Big Four endorsement or methodology replacement	Use "may use as discussion/read iness structure."
16 Conclusion	GAIC-SOURCE, AUD-01, AUD-03, AI-01	Object chain, audit evidence, assurance boundary, governance context	Main text	Insurability overclaim before WP3	Keep WP3 bridge high level.

Appendix Citation Map

Appendix	Source IDs used	Source role	Citation risk	R4 review note
A Evidence Request List	AUD-01, AUD-02, AUD-04, AUD-05, AUD-06, GAIC- SOURCE	Evidence/control vocabulary and MRO mapping	Formal audit- procedure implication	Keep request list as readiness architecture.
B Walkthrough Template	AUD-04, AUD-05, AUD-06, EVID-01, EVID-02, GAIC- SOURCE	Controls, provenance, traces, MRO lifecycle	Audit procedure implication	Keep illustrative.
C MRO Mapping	GAIC-SOURCE, AUD-01, AUD-05,	GAIC MRO source truth and audit/control	Legal mandate overclaim	Verify MRO

	AI-01	language		names exactly.
D Readiness Matrix	AUD-01, AUD-05, AUD-06, AI-01, BOUND-03, GAIC-SOURCE	Evidence, audit practice, governance, boundary	Certification/ maturity-score overclaim	Keep no scores.
E Closure Checklist	EVID-04, AI-02, AUD-04, AUD-05, AUD-06, GAIC-SOURCE	Incident, control, remediation, MRO closure	Legal closure overclaim	Closure means evidence state only.
F Boundary Language	AUD-03, BOUND-01, BOUND-02, BOUND-03, GAIC-SOURCE	Assurance, conformity, attestation, Validation Lab and companion-paper boundary	Boundary wording becoming legal disclaimer overreach	Keep as editorial guardrail.

R4 Citation Map Result

PASS - R4 candidate source uses R0 source IDs and GAIC source truth with explicit source-role and citation-risk notes.